

Samsung Biologics Co., Ltd.

Financial Statements

December 31, 2019

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Independent Auditor's Report

(English Translation of a Report Originally Issued in Korean)

To the Board of Directors and Shareholders of
Samsung Biologics Co., Ltd.

Opinion

We have audited the accompanying financial statements of Samsung Biologics Co., Ltd. (the "Company"), which comprise the statement of financial position as at December 31, 2019, and the statement of comprehensive income, statement of changes in equity and statement of cash flows for the year then ended, and notes to the financial statements, including a summary of significant accounting policies.

In our opinion, the accompanying financial statements present fairly, in all material respects, the financial position of Samsung Biologics Co., Ltd. as at December 31, 2019, and its financial performance and its cash flows for the year then ended in accordance with International Financial Reporting Standards as adopted by the Republic of Korea (Korean IFRS).

We also have audited, in accordance with Korean Standards on Auditing, the Company's Internal Control over Financial Reporting as at December 31, 2019, based on *Conceptual Framework for Designing and Operating Internal Control over Financial Reporting*, and our report dated March 5, 2020 expressed an unqualified opinion.

Basis for Opinion

We conducted our audit in accordance with Korean Standards on Auditing. Our responsibilities under those standards are further described in the *Auditor's Responsibilities for the Audit of the Financial Statements* section of our report. We are independent of the Company in accordance with the ethical requirements of the Republic of Korea that are relevant to our audit of the financial statements and we have fulfilled our other ethical responsibilities in accordance with these requirements. We believe that the audit evidence we have obtained is sufficient and appropriate to provide a basis for our audit opinion.

Emphasis of Matter

We draw attention to Note 35 to the financial statements.

Administrative lawsuit against a ruling of the Securities and Futures Commissions to suspend sanctions, and the related prosecution investigation and others

As described in Note 35 to the financial statements, the Securities and Futures Commission (the "SFC") made a decision to impose measures on the Company in relation to alleged errors in accounting for the Company's investment in Samsung Bioepis from 2012 to 2018.

The Company submitted requests to the Seoul Administrative Court (the "Court") to suspend the measures imposed by the SFC and the Court accepted the Company's requests. In relation to this, the appeal and re-appeal made by SFC were rejected by the Seoul High Court and the Supreme Court, respectively.

An administrative lawsuit and prosecution investigation against the Company's CEO and CFO are in progress, and it is not possible to predict the outcome of the administrative lawsuit and the investigation. However, if the Company loses the administrative lawsuit, then the Company's financial statements for 2012 and the subsequent reporting periods could potentially be restated as described in Note 35 to the financial statements.

Key Audit Matters

Key audit matters are those matters that, in our professional judgment, were of most significance in our audit of the financial statements of the current period. These matters were addressed in the context of our audit of the financial statements as a whole, and in forming our opinion thereon, and we do not provide a separate opinion on these matters.

(1) Impairment test of investments in associates and joint ventures

Why it is determined to be a Key Audit Matter

The Company accounts for investments in associates which the Company has significant influence and joint ventures which the Company holds joint control using the equity method of accounting. The Company reviews whether there are any impairment indicators of investments in associates and joint ventures at the end of each reporting period, and if there is an indication of impairment, the Company calculates the amount of impairment as the difference between the recoverable amount of investments in associates and joint ventures and their carrying value (Note 2). Investments in associates and joint ventures consist mostly of the Company's investment in Samsung Bioepis Co, Ltd. of ₩ 2,650 billion as of December 31, 2019. We determined that the impairment assessment of investments in associates and joint ventures to be a key audit matter as the investment in Samsung Bioepis Co., Ltd. comprise a material portion of the Company's financial statements and the determination of the recoverable amount involves significant judgement and estimates of the Company management including the determination of discount rates, revenue growth rates and cash flow forecast.

How our audit addressed the Key Audit Matter

Audit procedures we have performed in relation to the impairment assessment of the investment in Samsung Bioepis Co., Ltd. are as follows.

- Assessed the independence and competence of the external valuation specialists engaged by the Company
- Engaged our internal specialists, to assist in assessing the appropriateness of valuation method and key assumptions
- Assessed the reasonableness of the key assumptions and variables used in the assessment by comparing the latest business plans adopted by management with historical performance and available external data
- Compared the actual results for the current year with management's estimates in their cash flows calculations from the previous year in order to assess the historical accuracy of the management's forecasting process
- Tested mathematical accuracy of calculations and sensitivity analysis
- Assessed the adequacy of design and operation of management's internal controls.

(2) Revenue recognition of CMO

Why it is determined to be a Key Audit Matter

The Company operates a CMO(Contract Manufacturing Organization) business which is responsible for the production of bio-medicine. The Company's revenue recognition process primarily includes technology transfer procedures, regulatory approval procedures, and commercial production procedures. The Company recognizes revenue over the period of time as the control over produced bio-medicine transfers to the customers (Note 2). We determined that revenue recognition of the CMO to be a key audit matter as the revenue of the CMO business comprises a material portion of the Company's financial statements and the determination of the amount and timing of revenue recognition involves significant judgement and estimates of management.

How our audit addressed the Key Audit Matter

Audit procedures we have performed in relation to the CMO revenue recognition are as follows.

- Evaluated internal controls over the determination of the amount and timing of revenue recognition
- Inspected contracts for a sample of projects to assess the appropriateness of revenue recognition accounting
- Assessed the appropriateness of the determination of the amount and revenue recognition point through audit sampling.
- Performed independent recalculation of progress by project and contract assets and contract liabilities through audit sampling.

(3) Possibility of misstatement of the financial statements due to an inappropriate modification or deletion of major accounting records

Why it is determined to be a Key Audit Matter

Certain executives and employees of the Company and the Company's affiliates were found guilty of violating the Act on External Audit of Stock companies and Criminal Act in the first ruling of the Seoul Central District Court for forgery, destruction and concealment of evidence. The prosecutor's investigations for certain executives and employees are still ongoing (Note 35).

The Company formed an independent investigation committee who identified facts, determined whether there were any additional illegal acts or instances of fraud, and assessed the possibility of related misstatement of financial statements due to an inappropriate modification or deletion of major accounting records and deficiency in internal controls.

We identified this as a key audit matter because the impact on the deficiency in internal controls and the potential financial statement impact could be material.

How our audit addressed the Key Audit Matter

Audit procedures we have performed to assess whether the Company has inappropriately modified or deleted major accounting records and the possibility of related misstatement of financial statements are as follows.

- Assessed the independence and competence of the external expert that assisted investigation

committee in conducting the investigation.

- Assessed the adequacy of the scope and methodologies used by the investigation committee and assessed the impact of identified findings on the internal controls and the financial statements.
- Tested the completeness of the journal entries and financial data recorded in the financial system and whether such information have been modified or deleted inappropriately through an audit sampling.
- Assessed management's judgments and assumptions related to significant accounting estimates recognized in the beginning balance of the current year.
- Assessed the results of the investigation of certain matters at the Company's associate and potential impacts on the Company's financial statements.

Other Matters

The financial statements of the Company for the year ended December 31, 2018, were audited by KPMG Samjong Accounting Corp. who expressed an unqualified opinion on those statements on March 14, 2019 (however, the audit opinion on the statements of cash flows was dated July 23, 2019).

Auditing standards and their application in practice vary among countries. The procedures and practices used in the Republic of Korea to audit such financial statements may differ from those generally accepted and applied in other countries.

Responsibilities of Management and Those Charged with Governance for the Financial Statements

Management is responsible for the preparation and fair presentation of the financial statements in accordance with Korean IFRS, and for such internal control as management determines is necessary to enable the preparation of financial statements that are free from material misstatement, whether due to fraud or error.

In preparing the financial statements, management is responsible for assessing the Company's ability to continue as a going concern, disclosing, as applicable, matters related to going concern and using the going concern basis of accounting unless management either intends to liquidate the Company or to cease operations.

Those charged with governance are responsible for overseeing the Company's financial reporting process.

Auditor's Responsibilities for the Audit of the Financial Statements

Our objectives are to obtain reasonable assurance about whether the financial statements as a whole are free from material misstatement, whether due to fraud or error, and to issue an auditor's report that includes our opinion. Reasonable assurance is a high level of assurance, but is not a guarantee that an audit conducted in accordance with Korean Standards on Auditing will always detect a material misstatement when it exists. Misstatements can arise from fraud or error and are considered material if, individually or in the aggregate, they could reasonably be expected to influence the economic decisions of users taken on the basis of these financial statements.

As part of an audit in accordance with Korean Standards on Auditing, we exercise professional judgment and maintain professional skepticism throughout the audit. We also:

- Identify and assess the risks of material misstatement of the financial statements, whether due to fraud or error, design and perform audit procedures responsive to those risks, and obtain audit evidence that is sufficient and appropriate to provide a basis for our opinion. The risk of not detecting a material misstatement resulting from fraud is higher than for one resulting from error, as fraud may involve collusion, forgery, intentional omissions, misrepresentations, or the override of internal control.
- Obtain an understanding of internal control relevant to the audit in order to design audit procedures that are appropriate in the circumstances.
- Evaluate the appropriateness of accounting policies used and the reasonableness of accounting estimates and related disclosures made by management.
- Conclude on the appropriateness of management's use of the going concern basis of accounting and, based on the audit evidence obtained, whether a material uncertainty exists related to events or conditions that may cast significant doubt on the Company's ability to continue as a going concern. If we conclude that a material uncertainty exists, we are required to draw attention in our auditor's report to the related disclosures in the financial statements or, if such disclosures are inadequate, to modify our opinion. Our conclusions are based on the audit evidence obtained up to the date of our auditor's report. However, future events or conditions may cause the Company to cease to continue as a going concern.
- Evaluate the overall presentation, structure and content of the financial statements, including the disclosures, and whether the financial statements represent the underlying transactions and events in a manner that achieves fair presentation.

We communicate with those charged with governance regarding, among other matters, the planned scope and timing of the audit and significant audit findings, including any significant deficiencies in internal control that we identify during our audit.

We also provide those charged with governance with a statement that we have complied with relevant ethical requirements regarding independence, and to communicate with them all relationships and other matters that may reasonably be thought to bear on our independence, and where applicable, related safeguards.

From the matters communicated with those charged with governance, we determine those matters that were of most significance in the audit of the financial statements of the current period and are therefore the key audit matters. We describe these matters in our auditor's report unless law or regulation precludes public disclosure about the matter or when, in extremely rare circumstances, we determine that a matter should not be communicated in our report because the adverse consequences of doing so would reasonably be expected to outweigh the public interest benefits of such communication.

The engagement partner on the audit resulting in this independent auditor's report is Joonki Hong, Certified Public Accountant.

Samuel PricewaterhouseCoopers

Seoul, Korea

March 5, 2020

This report is effective as of March 5, 2020, the audit report date. Certain subsequent events or circumstances, which may occur between the audit report date and the time of reading this report, could have a material impact on the accompanying financial statements and notes thereto. Accordingly, the readers of the audit report should understand that there is a possibility that the above audit report may have to be revised to reflect the impact of such subsequent events or circumstances, if any.

Samsung Biologics Co., Ltd.
Statements of Financial Position
December 31, 2019 and 2018

<i>(in Korean won)</i>	Notes	2019	2018
Assets			
Current assets			
Cash and cash equivalents	5,6,34	₩ 125,100,653,179	₩ 17,621,904,942
Short-term financial instruments	5,34	320,000,000,000	550,000,000,000
Financial assets at fair value through profit or loss	5,34	210,040,293,295	540,080,516,898
Trade and other receivables	5,8,32,34	187,118,115,434	121,733,349,966
Inventories	10,26	452,469,927,424	302,360,715,341
Other current financial assets	5,11,34	34,782,011,074	20,879,609,446
Other current assets	2,12	5,938,312,588	5,508,442,675
Contract assets	9,34	20,812,351,177	19,499,176,306
		<u>1,356,261,664,171</u>	<u>1,577,683,715,574</u>
Non-current assets			
Long-term financial instruments	5,7,34	7,000,000	7,000,000
Investments in associates and joint ventures	2,13	2,652,731,622,024	2,561,802,520,579
Investment properties	14	20,761,092,845	-
Property, plant and equipment	15	1,702,792,524,097	1,717,720,987,857
Intangible assets	16	21,827,515,338	19,981,633,701
Right-of-use asset	2,20	16,118,883,277	-
Other non-current financial assets	5,11,34	9,498,952,389	24,230,355,452
Other non-current assets	2,12	956,721,462	1,902,973,235
Contract assets	9,34	130,670,536,317	77,079,461,895
		<u>4,555,364,847,749</u>	<u>4,402,724,932,719</u>
Total assets		<u>₩ 5,911,626,511,920</u>	<u>₩ 5,980,408,648,293</u>
Liabilities			
Current liabilities			
Trade and other payables	5,18,22,32,34	₩ 101,128,061,254	₩ 170,338,294,181
Emission liability	19	553,852,695	1,493,981,621
Debentures and borrowings	5,17,31,33,34	401,650,000,000	386,583,354,646
Other current liabilities	9,21,32	144,706,770,419	87,695,613,804
Contract liabilities	9,32	26,360,570,132	24,794,824,854
Lease liabilities	2,20,34	3,248,590,174	-
Current tax liabilities		12,857,398,029	39,623,717,556
		<u>690,505,242,703</u>	<u>710,529,786,662</u>

Samsung Biologics Co., Ltd.
Statements of Financial Position
December 31, 2019 and 2018

Non-current liabilities

Trade and other payables	18,22,32,34	8,105,040,694	3,988,729,379
Debentures and borrowings	5,17,31,33,34	236,167,124,961	486,112,751,634
Other non-current liabilities	9,21,32	6,011,768,076	-
Derivative liabilities	5	-	190,330,071
Defined benefit liabilities, net	22	14,429,295,289	13,266,305,086
Contract liabilities	9,32	118,768,686,348	76,309,964,692
Lease liabilities	2,20,34	12,640,693,837	-
Deferred tax liabilities, net	2,29	470,544,914,646	534,542,329,621

866,667,523,851 1,114,410,410,483

Total liabilities

1,557,172,766,554 1,824,940,197,145

Equity

Share capital	1,23	165,412,500,000	165,412,500,000
Share premium	23	2,487,313,082,024	2,487,313,082,024
Accumulated other comprehensive loss	13,22,23,29	(10,894,765,956)	(6,975,961,086)
Retained earnings	2,23,29	1,712,622,929,298	1,509,718,830,210

Total equity

4,354,453,745,366 4,155,468,451,148

Total liabilities and equity

₩ 5,911,626,511,920 ₩ 5,980,408,648,293

The above statements of financial position should be read in conjunction with the accompanying notes.

Samsung Biologics Co., Ltd.
Statements of Comprehensive Income
Years Ended December 31, 2019 and 2018

<i>(in Korean won)</i>	Note	2019	2018
Revenue	4,24,32	₩ 701,591,859,755	₩ 535,805,801,067
Cost of revenue	9,10,24,26,32	<u>(509,584,425,113)</u>	<u>(390,645,859,406)</u>
Gross profit		<u>192,007,434,642</u>	<u>145,159,941,661</u>
Selling, general and administrative expenses	25,26,32	<u>(100,265,047,444)</u>	<u>(89,481,197,135)</u>
Operating profit		<u>91,742,387,198</u>	<u>55,678,744,526</u>
Other income	27,32	4,003,486,749	390,399,125,406
Other expenses	27	(2,108,781,313)	(8,748,272,773)
Finance income	5,28	27,540,743,550	19,320,789,463
Finance costs	2,5,28	(38,682,642,614)	(24,337,216,586)
Share of profit (loss) of associates and joint ventures	2,13	<u>72,940,115,693</u>	<u>(129,337,332,786)</u>
Profit before income tax		155,435,309,263	302,975,837,250
Income tax benefit (expense)	2,29	<u>47,468,789,825</u>	<u>(78,866,642,378)</u>
Profit for the year		<u>202,904,099,088</u>	<u>224,109,194,872</u>
Other comprehensive loss		(3,918,804,870)	(2,966,514,221)
<i>Items that will not be reclassified to profit or loss</i>			
Remeasurements of net defined benefit liabilities	22,23,29	(3,787,818,691)	(3,930,106,654)
Share of other comprehensive income (loss) of associates and joint ventures	13,23,29	37,307,714	(663,926,112)
<i>Items that may be subsequently reclassified to profit or loss</i>			
Share of other comprehensive income (loss) of associates and joint ventures	13,23,29	<u>(168,293,893)</u>	<u>1,627,518,545</u>
Total comprehensive income for the year		<u>₩ 198,985,294,218</u>	<u>₩ 221,142,680,651</u>
Earnings per share			
Basic and diluted earnings per share	30	₩ 3,067	₩ 3,387

The above statements of comprehensive income should be read in conjunction with the accompanying notes.

Samsung Biologics Co., Ltd.
Statements of Changes in Equity
Years Ended December 31, 2019 and 2018

(in Korean won)

	Notes	Share capital	Share premium	Accumulated other comprehensive loss	Retained earnings	Total
Balance at January 1, 2018		₩ 165,412,500,000	₩ 2,487,313,082,024	₩ (4,009,446,865)	₩ 1,327,734,757,202	₩ 3,976,450,892,361
Adjustment on initial application of Korean IFRS 1115 (after tax)		-	-	-	(41,974,908,757)	(41,974,908,757)
Restated at January 1, 2018		165,412,500,000	2,487,313,082,024	(4,009,446,865)	1,285,759,848,445	3,934,475,983,604
Total comprehensive income (loss)						
Profit for the year		-	-	-	224,109,194,872	224,109,194,872
Other comprehensive income (loss)						
Remeasurements of net defined benefit liabilities	22,23	-	-	(3,930,106,654)	-	(3,930,106,654)
Share of other comprehensive income of associates and joint ventures	13,23	-	-	640,127,389	-	640,127,389
Disposal of investments in associates and joint ventures	13	-	-	323,465,044	(150,213,107)	173,251,937
Total comprehensive income (loss) for the year		-	-	(2,966,514,221)	223,958,981,765	220,992,467,544
Balance at December 31, 2018		₩ 165,412,500,000	₩ 2,487,313,082,024	₩ (6,975,961,086)	₩ 1,509,718,830,210	₩ 4,155,468,451,148
Balance at January 1, 2019		₩ 165,412,500,000	₩ 2,487,313,082,024	₩ (6,975,961,086)	₩ 1,509,718,830,210	₩ 4,155,468,451,148
Total comprehensive loss						
Profit for the year		-	-	-	202,904,099,088	202,904,099,088
Other comprehensive loss						
Remeasurements of net defined benefit liabilities	22,23	-	-	(3,787,818,691)	-	(3,787,818,691)
Share of other comprehensive loss of associates and joint ventures	13,23	-	-	(130,986,179)	-	(130,986,179)
Total comprehensive income (loss) for the year		-	-	(3,918,804,870)	202,904,099,088	198,985,294,218
Balance at December 31, 2019		₩ 165,412,500,000	₩ 2,487,313,082,024	₩ (10,894,765,956)	₩ 1,712,622,929,298	₩ 4,354,453,745,366

The above statements of changes in equity should be read in conjunction with the accompanying notes.

Samsung Biologics Co., Ltd.
Statements of Cash Flows
Years Ended December 31, 2019 and 2018

<i>(in Korean won)</i>	Notes	2019	2018
Cash flows from operating activities			
Profit for the year	₩	202,904,099,088	₩ 224,109,194,872
Adjustments for:			
Retirement benefits	22	14,710,433,760	10,070,883,785
Depreciation of investment properties	14	342,088,125	-
Depreciation	15,25,26	121,892,350,206	85,522,361,717
Amortization of intangible assets	16,25,26	6,292,084,706	5,928,193,275
Depreciation of right-of-use asset	20	4,358,579,512	-
(Reversal of) Loss on valuation of inventories	10	6,215,382,569	(1,932,844,118)
(Reversal of) Impairment loss on contract assets	9	(2,263,140,740)	18,641,061,158
Loss on retirement of property, plant and equipment	27	296,624,745	-
Gain on disposal of property, plant and equipment	27	(520,162)	(12,821,049)
Loss on disposal of property, plant and equipment	27	312,536,821	614,509,074
Loss on disposal of intangible assets	27	197,566,643	-
Impairment loss on intangible assets	27	359,600,000	10,756,000
Gain on foreign currency translations	28	(143,426,108)	(112,645,616)
Loss on foreign currency translations	28	7,436,126,539	3,291,913,084
Gain on valuation of derivatives	5,28	(1,193,320,503)	-
Loss on valuation of derivatives	5,28	-	190,330,071
Cost of emissions	19	553,852,695	1,493,981,621
Interest income	5,28	(17,246,463,926)	(9,651,185,568)
Interest expenses	5,28	25,614,721,779	15,838,589,038
Gain on valuation of financial assets at fair value through profit or loss	5,28	(45,759,047)	(80,516,898)
Gain on disposal of financial assets at fair value through profit or loss	5,28	(361,458,825)	(301,926,649)
Share of loss (profit) of associates and joint ventures	13	(72,940,115,693)	129,337,332,786
Gain on disposal of investments in associates and joint ventures	27	-	(389,213,671,963)
Income tax expense (benefit)	29	(47,468,789,825)	78,866,642,378
Others		41,349,843	(45,800,000)
		<u>46,960,303,114</u>	<u>(51,544,857,874)</u>
Change in operating assets and liabilities:			
Trade and other receivables		(69,000,556,074)	(96,391,796,775)
Other current assets		(270,600,862)	(368,023,369)
Inventories		(156,324,594,652)	(90,441,154,623)
Deposits provided		2,471,171,102	(495,032,437)
Other non-current assets		(578,825,299)	(1,598,129,283)
Contract assets		(52,659,037,559)	(73,070,459,350)
Trade and other payables		(150,251,947)	3,064,884,544
Other current liabilities		63,029,175,650	31,209,439,258
Other non-current payables		1,611,419,158	2,863,550,910
Contract liabilities		44,024,466,934	43,664,407,237

Samsung Biologics Co., Ltd.
Statements of Cash Flows
Years Ended December 31, 2019 and 2018

Emission liability	19	(1,493,981,621)	(502,080,446)
Retirement benefits paid	22	(1,358,362,970)	(906,052,780)
Transfer from affiliates of defined benefit liabilities	22	-	30,574,353
Contributions to plan assets	22	(17,186,202,871)	(6,638,266,129)
		<u>(187,886,181,011)</u>	<u>(189,578,138,890)</u>
Interest received		16,625,472,877	8,486,910,234
Interest paid		(26,345,466,077)	(23,359,097,778)
Income tax paid		<u>(42,104,613,015)</u>	<u>(1,131,337,889)</u>
Net cash inflow (outflow) from operating activities		<u>10,153,614,976</u>	<u>(33,017,327,325)</u>
Cash flows from investing activities			
Decrease in short-term financial instruments		3,280,000,000,000	2,730,000,000,000
Decrease in financial assets at fair value through profit or loss		1,600,000,000,000	750,000,000,000
Decrease in investments in associates and joint ventures		-	759,460,611,885
Disposal of property, plant and equipment		42,326,492	13,207,800
Increase in short-term financial instruments		(3,050,000,000,000)	(3,280,000,000,000)
Increase in financial assets at fair value through profit or loss		(1,270,000,000,000)	(1,290,000,000,000)
Acquisition of investment properties		(21,103,180,970)	-
Acquisition of property, plant and equipment		(171,734,357,701)	(278,647,255,715)
Acquisition of intangible assets		(9,461,855,113)	(5,811,483,125)
Acquisition of investments in associates and joint ventures		<u>(18,101,000,000)</u>	<u>(16,405,000,000)</u>
Net cash inflow (outflow) from investing activities		<u>339,641,932,708</u>	<u>(631,389,919,155)</u>
Cash flows from financing activities			
Proceeds from debentures issued	17,33	-	189,579,060,000
Proceeds from short-term borrowings	17,33	150,000,000,000	50,000,000,000
Proceeds from long-term borrowings	17,33	-	153,112,000,000
Repayment of debentures	17,33	(50,000,000,000)	(50,000,000,000)
Repayment of short-term borrowings	17,33	(90,000,000,000)	-
Repayment of lease liabilities	20,33	(3,222,219,801)	-
Repayment of current portion of long-term borrowings	17,33	<u>(246,583,354,646)</u>	<u>(18,750,000,000)</u>
Net cash inflow (outflow) from financing activities		<u>(239,805,574,447)</u>	<u>323,941,060,000</u>
Net increase (decrease) in cash and cash equivalents		109,989,973,237	(340,466,186,480)
Cash and cash equivalents at the beginning of the year		17,621,904,942	358,088,091,422
Effects of exchange rate changes on cash and cash equivalents		<u>(2,511,225,000)</u>	<u>-</u>
Cash and cash equivalents at the end of the year		<u>W 125,100,653,179</u>	<u>W 17,621,904,942</u>

The above statements of cash flows should be read in conjunction with the accompanying notes.

Samsung Biologics Co., Ltd.

Notes to the Financial Statements

December 31, 2019 and 2018

1. General Information

Samsung Biologics Co., Ltd. (the "Company") was established on April 22, 2011 under the commercial law of the Republic of Korea and is engaged in Biosimilar manufacturing. The Company's head office is located in 300, Songdobio-daero, Yeonsu-gu, Incheon-si, The Republic of Korea. The Company was listed on securities market on November 10, 2016. As at December 31, 2019, the Company's share capital is ₩ 165,413 million.

The shareholders of the Company as at December 31, 2019, are as follows:

Shareholder	Number of shares	Percentage of ownership (%)
Samsung C&T Corporation	28,742,466	43.44
Samsung Electronics Co., Ltd.	20,836,832	31.49
Others	16,585,702	25.07
	<u>66,165,000</u>	<u>100.00</u>

2. Significant Accounting Policies

The principal accounting policies applied in the preparation of these financial statements are set out below. These policies have been consistently applied to all the years presented, unless otherwise stated.

2.1 Basis of Preparation

The Company maintains its accounting records in Korean won and prepares statutory financial statements in the Korean language (Hangul) in accordance with International Financial Reporting Standards as adopted by the Republic of Korea (Korean IFRS). The accompanying financial statements have been condensed, restructured and translated into English from the Korean language financial statements.

Certain information attached to the Korean language financial statements, but not required for a fair presentation of the Company's financial position, financial performance or cash flows, is not presented in the accompanying financial statements.

The financial statements of the Company have been prepared in accordance with Korean IFRS. These are the standards, subsequent amendments and related interpretations issued by the International Accounting Standards Board (IASB) that have been adopted by the Republic of Korea.

The financial statements have been prepared on a historical cost basis, except for the following:

- Certain financial assets and liabilities (including derivative instruments), and
- Defined benefit pension plans – plan assets measured at fair value.

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The preparation of financial statements requires the use of critical accounting estimates. Management also needs to exercise judgement in applying the Company's accounting policies. The areas involving a higher degree of judgment or complexity, or areas where assumptions and estimates are significant to the financial statements are disclosed in Note 3.

2.2 Changes in Accounting Policies and Disclosures

2.2.1 New and amended standards adopted by the Company

The Company has applied the following standards and amendments for the first time for their annual reporting period commencing January 1, 2019.

(a) Enactment of Korean IFRS 1116 Leases

Under the new standard, with implementation of a single lease model, lessee is required to recognize assets and liabilities for all lease which lease term is over 12 months and underlying assets are not low value assets. A lessee is required to recognize a right-of-use asset and a lease liability representing its obligation to make lease payments.

The Company has adopted Korean IFRS 1116 retrospectively, as permitted under the specific transitional provisions in the standard, and recognized the cumulative impact of initially applying the standard as at January 1, 2019, the date of initial application. The Company has not restated comparatives for the 2018 reporting period.

The impact of the adoption of the leasing standard and the new accounting policies are as follows.

i) Adjustments recognized on adoption of Korean IFRS 1116 Lease

On adoption of Korean IFRS 1116, the Company recognized right-of-use assets and lease liabilities in relation to leases which had previously been classified as 'operating leases' under Korean IFRS 1017. These liabilities were measured at the present value of the remaining lease payments, discounted using the lessee's incremental borrowing rate as at January 1, 2019. The weighted average lessee's incremental borrowing rate applied to the lease liabilities on January 1, 2019 was 3.5%.

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(in thousands of Korean won)

2019

Operating lease commitments	₩	20,007,710
Discounted using the lessee's incremental borrowing rate at the date of initial application		18,010,132
Low-value leases recognized on a straight-line basis as expense		(232,794)
Lease liability recognized as at January 1, 2019		17,777,338
Of which are:		
Current lease liabilities		3,332,778
Non-current lease liabilities		14,444,560
	₩	17,777,338

ii) Amounts recognized in the statement of financial position

There is no impact on retained earnings on January 1, 2019.

Impact on the statement of financial position as at December 31, 2019, is as follows:

(in thousands of Korean won)	As reported	Adjustments	Amounts without adoption of Korean IFRS 1116
Current assets:			
Prepaid expenses	₩ 5,919,923	₩ 277,412	₩ 6,197,335
Non-current assets:			
Prepaid expenses	956,721	216,507	1,173,228
Right-of-use assets	16,118,883	(16,118,883)	-
Investments in associates and joint ventures	2,652,731,622	202,794	2,652,934,416
Current liabilities:			
Lease liabilities	3,248,590	(3,248,590)	-
Non-current liabilities:			
Lease liabilities	12,640,694	(12,640,694)	-
Deferred tax liabilities	470,544,914	63,965	470,608,879
Equity:			
Retained earnings	1,712,622,929	403,149	1,713,026,078

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Impact on the statement of comprehensive income for the year ended December 31, 2019, is as follows:

<i>(in thousands of Korean won)</i>	As reported	Adjustments	Amounts without adoption of Korean IFRS 1116
Revenue	₩ 701,591,860	₩ -	₩ 701,591,860
Cost of revenue	(509,584,425)	-	(509,584,425)
Gross profit	192,007,435	-	192,007,435
Operating profit (loss)	91,742,387	(256,901)	91,485,486
Other income	4,003,487	(117)	4,003,370
Other expenses	(2,108,781)	-	(2,108,781)
Finance income	27,540,744	-	27,540,744
Finance costs	(38,682,643)	521,337	(38,161,306)
Share of profit of associates and joint ventures	72,940,116	202,794	73,142,910
Profit before income tax	155,435,310	467,114	155,902,424
Income tax benefit (expense)	47,468,790	(63,965)	47,404,825
Profit for the year	202,904,100	403,149	203,307,249

(b) Amendment to Korean IFRS 1109 Financial Instruments – Prepayment Features with Negative Compensation

The narrow-scope amendments made to Korean IFRS 1109 *Financial Instruments* enable entities to measure certain prepayable financial assets with negative compensation at amortized cost. When a modification of a financial liability measured at amortized cost that does not result in the derecognition, a modification gain or loss shall be recognized in profit or loss. The amendment does not have a significant impact on the financial statements.

(c) Amendments to Korean IFRS 1019 Employee Benefits

The amendments require that an entity shall calculate current service cost and net interest for the remainder of the reporting period after a plan amendment, curtailment or settlement based on updated actuarial assumptions from the date of the change. The amendments also require that a reduction in a surplus must be recognized in profit or loss even if that surplus was not previously recognized because of the impact of the asset ceiling. The amendment does not have a significant impact on the financial statements.

(d) Amendments to Korean IFRS 1028 Investments in Associates and Joint Ventures

The amendments clarify that an entity shall apply Korean IFRS 1109 to financial instruments in an associate or joint venture to which the equity method is not applied. The amendments also clarify that Korean IFRS 1109 requirements are applied to long-term interests that form part of the entity's net investment in an associate or joint venture before applying the impairment requirements of Korean IFRS 1028. The amendment does not have a significant impact on the

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financial statements.

(e) Enactment to Interpretation of Korean IFRS 2123 *Uncertainty over Income Tax Treatments*

The interpretation explains how to recognize and measure deferred and current income tax assets and liabilities where there is uncertainty over a tax treatment, and includes guidance on how to determine whether each uncertain tax treatment is considered separately or together. It also presents examples of circumstances where a judgement or estimate is required to be reassessed. The enactment does not have a significant impact on the financial statements.

(f) Annual Improvements to Korean IFRS 2015 – 2017 Cycle:

- Amendments to Korean IFRS 1103 *Business Combination*

The amendments clarify that when a party to a joint arrangement obtains control of a business that is a joint operation, and had rights to the assets and obligations for the liabilities relating to that joint operation immediately before the acquisition date, the transaction is a business combination achieved in stages. In such cases, the acquirer shall remeasure its entire previously held interest in the joint operation. The amendment does not have a significant impact on the financial statements.

- Amendments to Korean IFRS 1111 *Joint Agreements*

The amendments clarify that when a party that participates in, but does not have joint control of, a joint operation might obtain joint control of the joint operation in which the activity of the joint operation constitutes a business. In such cases, previously held interests in the joint operation are not remeasured. The amendment does not have a significant impact on the financial statements.

- Amendments to Paragraph 57A of Korean IFRS 1012 *Income Tax*

The amendment is applied to all the income tax consequences of dividends and requires an entity to recognize the income tax consequences of dividends in profit or loss, other comprehensive income or equity according to where the entity originally recognized those past transactions or events. The amendment does not have a significant impact on the financial statements.

- Korean IFRS 1023 *Borrowing Costs*

The amendments clarify that if a specific borrowing remains outstanding after the related qualifying asset is ready for its intended use (or sale), it becomes part of general borrowings. The amendment does not have a significant impact on the financial statements.

2.2.2 New standards and interpretations not yet adopted by the Company

The following new accounting standards and interpretations have been published that are not mandatory for December 31, 2019 reporting periods and have not been early adopted by the Company.

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(a) Amendments to Korean IFRS 1001 Presentation of Financial Statements and Korean IFRS 1008 Accounting policies, changes in accounting estimates and errors – Definition of Material

The amendments clarify the explanation of the definition of material and amended Korean IFRS 1001 and Korean IFRS 1008 in accordance with the clarified definitions. Materiality is assessed by reference to omission or misstatement of material information as well as effects of immaterial information, and to the nature of the users when determining the information to be disclosed by the Company. These amendments should be applied for annual periods beginning on or after January 1, 2020, and earlier application of permitted. The Company does not expect that these amendments have a significant impact on the financial statements.

(b) Amendments to Korean IFRS 1103 Business Combination – Definition of a Business

To consider the integration of the required activities and assets as a business, the amended definition of a business requires an acquisition to include an input and a substantive process that together significantly contribute to the ability to create outputs and excludes economic benefits from the lower costs. An entity can apply a concentration test, an optional test, where substantially all of the fair value of gross assets acquired is concentrated in a single asset or a group of similar assets, the assets acquired would not represent a business. These amendments should be applied for annual periods beginning on or after January 1, 2020, and earlier application of permitted. The Company does not expect that these amendments have a significant impact on the financial statements.

2.3 Cash and Cash Equivalents Financial Assets

Cash and cash equivalents include cash on hand, deposits held at call with banks, and other short-term highly liquid investments that are readily convertible to a known amount of cash and are subject to an insignificant risk of change in value.

2.4 Associates and Joint Ventures

Associates are those entities in which the Company has significant influence, but no control or joint control, over the financial and operating policies.

A joint venture is an arrangement in which the Company has joint control, whereby the Company has rights to the net assets of the arrangement, rather than rights to its assets and obligations for its liabilities.

Investments in associates and joint ventures are accounted for using the equity method of accounting, after initially being recognized at cost. Unrealized gains on transactions between the Company and its associates and joint ventures are eliminated to the extent of the Company's interest in the associates and joint ventures. If the Company's share of losses of an associate and joint venture equals or exceeds its interest in the associate and joint venture (including long-term interests that, in substance, form part of the Company's net investment in the associate and joint venture), the Company discontinues recognizing its share of further losses. After the Company's interest is reduced to zero, additional losses are provided for, and a liability is recognized, only to

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the extent that the Company has incurred legal or constructive obligations or made payments on behalf of the associate. If there is an objective evidence of impairment for the investment in the associate and joint venture, the Company recognizes the difference between the recoverable amount of the associate and joint venture and its book amount as impairment loss. If an associate and joint venture uses accounting policies other than those of the Company for like transactions and events in similar circumstances, if necessary, adjustments shall be made to make the accounting policies of the associate and joint venture conform to those of the Company when the financial statements of the associate and joint venture are used by the Company in applying the equity method.

2.5 Foreign Currency Translation

(a) Functional and presentation currency

Items included in the financial statements of the Company are measured using the currency of the primary economic environment in which each entity operates (the “functional currency”). The financial statements are presented in Korean won, which is the Company’s functional and presentation currency.

(b) Transactions and balances

Foreign currency transactions are translated into the functional currency using the exchange rates at the dates of the transactions. Foreign exchange gains and losses resulting from the settlement of such transactions and from the translation of monetary assets and liabilities denominated in foreign currencies at year end exchange rates are generally recognized in profit or loss. They are deferred in other comprehensive income if they relate to qualifying cash flow hedges and qualifying effective portion of net investment hedges, or are attributable to monetary part of the net investment in a foreign operation.

Translation differences on assets and liabilities carried at fair value are reported as part of the fair value gain or loss. For example, translation differences on non-monetary assets and liabilities such as equities held at fair value through profit or loss are recognized in profit or loss as part of the fair value gain or loss and translation differences on non-monetary assets such as equities held at fair value through other comprehensive income are recognized in other comprehensive income.

2.6 Financial Assets

(a) Classification

The Company classifies its financial assets in the following measurement categories:

- those to be measured at amortized cost
- those to be measured at fair value through other comprehensive income, and
- those to be measured at fair value through profit or loss.

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The classification depends on the Company's business model for managing the financial assets and the contractual terms of the cash flows.

For financial assets measured at fair value, gains and losses will either be recorded in profit or loss or other comprehensive income. For investments in debt instruments, this will depend on the business model in which the investment is held. The Company reclassifies debt investments when, and only when its business model for managing those assets changes.

For investments in equity instruments that are not held for trading, this will depend on whether the Company has made an irrevocable election at the time of initial recognition to account for the equity investment at fair value through other comprehensive income. Changes in fair value of non-designated equity investment are recognized in profit or loss.

(b) Measurement

At initial recognition, the Company measures a financial asset at its fair value plus, in the case of a financial asset not at fair value through profit or loss, transaction costs that are directly attributable to the acquisition of the financial asset. Transaction costs of financial assets carried at fair value through profit or loss are expensed in profit or loss.

Financial assets with embedded derivatives are considered in their entirety when determining whether their cash flows are solely payment of principal and interest.

A. Debt instruments

Subsequent measurement of debt instruments depends on the Company's business model for managing the asset and the cash flow characteristics of the asset. The Company classifies its debt instruments into one of the following three measurement categories:

- **Amortized cost:** Assets that are held for collection of contractual cash flows where those cash flows represent solely payments of principal and interest are measured at amortized cost. A gain or loss on a debt investment that is subsequently measured at amortized cost and is not part of a hedging relationship is recognized in profit or loss when the asset is derecognized or impaired. Interest income from these financial assets is included in 'finance income' using the effective interest rate method.
- **Fair value through other comprehensive income:** Assets that are held for collection of contractual cash flows and for selling the financial assets, where the assets' cash flows represent solely payments of principal and interest, are measured at fair value through other comprehensive income. Movements in the carrying amount are taken through other comprehensive income, except for the recognition of impairment loss (reversal of impairment loss), interest income and foreign exchange gains and losses which are recognized in profit or loss. When the financial asset is derecognized, the cumulative gain or loss previously recognized in other comprehensive income is reclassified from equity to profit or loss.
- **Fair value through profit or loss:** Assets that do not meet the criteria for amortized cost or fair value through other comprehensive income are measured at fair value through profit or loss. A gain or loss on a debt investment that is subsequently measured at fair value through profit or loss and is not part of a hedging relationship is recognized in 'finance income or costs'.

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B. Equity instruments

The Company subsequently measures all equity investments at fair value. Where the Company's management has elected to present fair value gains and losses on equity investments, which held for long-term investment or strategic purpose, in other comprehensive income, there is no subsequent reclassification of fair value gains and losses to profit or loss following the derecognition of the investment. Dividend income from such investments continue to be recognized as 'finance income' when the right to receive payments is established.

Changes in the fair value of financial assets at fair value through profit or loss are recognized in 'finance income and expenses'. Impairment loss (reversal of impairment loss) on equity investments measured at fair value through other comprehensive income are not reported separately from other changes in fair value.

(c) Impairment

The Company assesses on a forward-looking basis the expected credit losses associated with its debt instruments carried at amortized cost and fair value through other comprehensive income. The impairment methodology applied depends on whether there has been a significant increase in credit risk. For trade receivables and lease receivables, the Company applies the simplified approach, which requires expected lifetime credit losses to be recognized from initial recognition of the receivables.

(d) Recognition and Derecognition

Regular way purchases and sales of financial assets are recognized or derecognized on trade-date, the date on which the Company commits to purchase or sell the asset. Financial assets are derecognized when the rights to receive cash flows from the financial assets have expired or have been transferred and the Company has transferred substantially all the risks and rewards of ownership.

If a transfer does not result in derecognition because the Company has retained substantially all the risks and rewards of ownership of the transferred asset, the Company continues to recognize the transferred asset in its entirety and recognizes a financial liability for the consideration received.

(e) Offsetting of financial instruments

Financial assets and liabilities are offset and the net amount reported in the statement of financial position where there is a legally enforceable right to offset the recognized amounts and there is an intention to settle on a net basis or realize the assets and settle the liability simultaneously. The legally enforceable right must not be contingent on future events and must be enforceable in the normal course of business and in the event of default, insolvency or bankruptcy of the Company or the counterparty.

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2.7 Derivative Instruments

Derivatives are initially recognized at fair value on the date when a derivative contract is entered into and are subsequently remeasured at their fair value at the end of each reporting period. Changes in the fair value of any derivative instrument that does not qualify for hedge accounting are recognized immediately in profit or loss based on the nature of transactions.

2.8 Trade Receivables

Trade receivables are amounts due from customers for merchandise sold or services performed in the ordinary course of business. Trade receivables are recognized initially at the amount of consideration that is unconditional, unless they contain significant financing components when they are recognized at fair value.

2.9 Inventories

Inventories are stated at the lower of cost and net realizable value. Cost is determined using the individual method or moving weighted average method. The cost of inventories includes expenditures for acquiring the inventories, production or conversion costs and other costs incurred in bringing them to their existing location and condition.

Inventories are measured at the lower of cost and net realizable value. The amount of any write-down of inventories to net realizable value and all losses of inventories are recognized as an expense in the period the write-down or loss occurs. The amount of any reversal of any write-down of inventories, arising from an increase in net realizable value, are recognized as a reduction in the amount of inventories recognized as cost of revenue in the period in which the reversal occurs.

2.10 Property, Plant and Equipment

Property, plant and equipment are stated at historical cost less accumulated depreciation and accumulated impairment losses.

Historical cost includes expenditure that is directly attributable to the acquisition of the items. Subsequent costs are included in the asset's carrying amount or recognized as a separate asset, as appropriate, only when it is probable that future economic benefits associated with the item will flow to the Company and the cost of the item can be measured reliably. The carrying amount of the replaced part is derecognized. All other repairs and maintenance are charged to the statements of other comprehensive income during the financial period in which they are incurred.

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Depreciation of all property, plant and equipment, except for land, is calculated using the straight-line method to allocate their cost or revalued amounts, net of their residual values, over their estimated useful lives as follows:

	Useful lives
Buildings and Structures	40 years
Machinery	5 – 15
Vehicles	4
Others	4

The assets' depreciation method, residual values and useful lives are reviewed, and adjusted if appropriate, at the end of each reporting period.

Gains and losses on disposal of an item of property, plant and equipment are determined by comparing the proceeds from disposal with the carrying amount of property, plant and equipment and are recognized in profit or loss.

2.11 Borrowing Costs

The Company capitalizes borrowing costs directly attributable to the acquisition, construction, or production of a qualifying asset as part of the cost of that asset. Other borrowing costs are recognized in expense as incurred. A qualifying asset is an asset that requires a substantial period of time to get ready for its intended use or sale. Financial assets and inventories that are manufactured or otherwise produced over a short period of time are not qualifying assets. Assets that are ready for their intended use or sale when acquired are not qualifying assets.

To the extent that the Company borrows funds specifically for the purpose of obtaining a qualifying asset, the Company determines the amount of borrowing costs eligible for capitalization as the actual borrowing costs incurred on that borrowing during the period less any investment income on the temporary investment of those borrowings. To the extent that the Company borrows funds generally and uses them for the purpose of obtaining a qualifying asset, the Company shall determine the amount of borrowing costs eligible for capitalization by applying a capitalization rate to the expenditures on that asset. The capitalization rate shall be the weighted average of the borrowing costs applicable to the borrowings of the Company that are outstanding during the period, other than borrowings made specifically for the purpose of obtaining a qualifying asset. The amount of borrowing costs that the Company capitalizes during a period shall not exceed the amount of borrowing costs incurred during that period.

2.12 Government Grants

Grants from the government are recognized at their fair value where there is a reasonable assurance that the grant will be received and the Company will comply with all attached conditions.

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In connection with a lease agreement with Incheon Metropolitan city ("Incheon") related to the use of land, the Company is exempt from the full amount of rent because the Company meets the requirements for exemption of rent in accordance with the Common Property Management Ordinance of Incheon. The Company recognized an intangible asset at the fair value of the rent that is exempt from Incheon and government grant as a reduction of assets at the same value.

2.13 Intangible Assets

Intangible assets are initially recognized at its historical cost, and carried at cost less accumulated amortization and accumulated impairment losses.

Subsequent expenditures are capitalized only when they increase the future economic benefits embodied in the specific asset to which it relates. All other expenditures, including expenditures on internally generated goodwill, are recognized in profit or loss as incurred.

Amortization of intangible assets is calculated on a straight-line basis over the following estimated useful lives of intangible assets from the date that they are available for use. The residual value of intangible assets is zero. However, as there are no foreseeable limits to the periods over which memberships are expected to be available for use, they are determined as having indefinite useful lives and not amortized.

	Useful lives
Software	5 years
Other intangible assets	5
Land use rights	50
Memberships	indefinite

Useful lives and the amortization methods for intangible assets with finite useful lives are reviewed at the end of each reporting period. The useful lives of intangible assets that are not being amortized are reviewed at the end of each reporting period to determine whether events and circumstances continue to support indefinite useful life assessments for those assets. Changes are accounted for as changes in accounting estimates.

2.14 Investment Property

Investment property is property held to earn rentals or for capital appreciation or both. An investment property is measured initially at its cost. An investment property is measured after initial measurement at depreciated cost (less any accumulated impairment losses). After recognition as an asset, investment property is carried at cost less accumulated depreciation and impairment losses. The Company depreciates investment properties, except for landscaping, using the straight-line method over their useful lives.

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2.15 Impairment of Non-financial Assets

Goodwill and intangible assets that have an indefinite useful life are not subject to amortization and are tested annually for impairment, or more frequently if events or changes in circumstances indicate that they might be impaired. Other assets are tested for impairment whenever events or changes in circumstances indicate that the carrying amount may not be recoverable. An impairment loss is recognized for the amount by which the asset's carrying amount exceeds its recoverable amount. The recoverable amount is the higher of an asset's fair value less costs of disposal and value in use. Non-financial assets other than goodwill that suffered an impairment are reviewed for possible reversal of the impairment at the end of each reporting period.

2.16 Trade and Other Payables

These amounts represent liabilities for goods and services provided to the Company prior to the end of reporting period which are unpaid. Trade and other payables are presented as current liabilities, unless payment is not due within 12 months after the reporting period.

2.17 Financial Liabilities

(a) Classification and measurement

The Company's financial liabilities at fair value through profit or loss are financial instruments held for trading. A financial liability is held for trading if it is incurred principally for the purpose of repurchasing in the near term. A derivative that is not a designated as hedging instruments and an embedded derivative that is separated are also classified as held for trading.

The Company classifies non-derivative financial liabilities, except for financial liabilities at fair value through profit or loss, financial guarantee contracts and financial liabilities that arise when a transfer of financial assets does not qualify for derecognition, as financial liabilities carried at amortized cost and present as 'trade and other payables' and 'debentures and borrowings' in the statement of financial position.

(b) Derecognition

Financial liabilities are removed from the statement of financial position when it is extinguished; for example, when the obligation specified in the contract is discharged or cancelled or expired or when the terms of an existing financial liability are substantially modified. The difference between the carrying amount of a financial liability extinguished or transferred to another party and the consideration paid (including any non-cash assets transferred or liabilities assumed) is recognized in profit or loss.

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2.18 Provisions and Contingencies

Provisions are recognized when the Company has a present legal or constructive obligation as a result of past events, it is probable that an outflow of resources will be required to settle the obligation and the amount can be reliably estimated. Provisions are measured at the present value of management's best estimate of the expenditure required to settle the present obligation at the end of the reporting period, and the increase in the provision due to the passage of time is recognized as interest expense.

A contingent liability is disclosed when; - a possible obligation that arises from past events and whose existence will be confirmed only by the occurrence or non-occurrence of one or more uncertain future events not wholly within the control of the entity; or - a present obligation that arises from past events but is not recognized because: it is not probable that an outflow of resources embodying economic benefits will be required to settle the obligation; or the amount of the obligation cannot be measured with sufficient reliability.

2.19 Greenhouse Gas Emission Permits

With enforcement of The Act on the Allocation and Trading of Greenhouse Gas Emission Permits, the allocation received from the government for free of charge are measured at zero while purchased emission permits are measured at acquisition cost and presented net of accumulated impairment loss. Emissions obligations are measured as the sum of the carrying amount of the allocated allowances that will be submitted to the government and the best estimate of expenditure required to settle the obligation at the end of reporting period for any excess emission. Emission permits and emission obligations are classified as intangible assets and emission liability, respectively, in the statement of financial position.

2.20 Current and Deferred Tax

The tax expense for the period consists of current and deferred tax. Current and deferred tax is recognized in profit or loss, except to the extent that it relates to items recognized in other comprehensive income or directly in equity. In this case, the tax is also recognized in other comprehensive income or directly in equity, respectively.

The current income tax expense is measured at the amount expected to be paid to the taxation authorities, using the tax rates (and tax laws) that have been enacted or substantively enacted by the end of the reporting period. Management periodically evaluates positions taken in tax returns with respect to situations in which applicable tax regulation is subject to interpretation. The Company recognizes current income tax on the basis of amounts expected to be paid to the tax authorities.

Deferred income tax is provided in full on temporary differences arising between the tax bases of assets and liabilities and their carrying amounts in the financial statements. However, deferred income tax is not accounted for if it arises from initial recognition of an asset or liability in a transaction other than a business combination that at the time of the transaction affects neither accounting profit nor taxable profit or loss.

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Deferred tax assets are recognized only if it is probable that future taxable amounts will be available to utilize those temporary differences and losses.

The Company recognizes a deferred tax liability all taxable temporary differences associated with investments in associates, and interests in joint arrangements, except to the extent that the Company is able to control the timing of the reversal of the temporary difference and it is probable that the temporary difference will not reverse in the foreseeable future. In addition, The Company recognizes a deferred tax asset for all deductible temporary differences arising from such investments to the extent that it is probable the temporary difference will reverse in the foreseeable future and taxable profit will be available against which the temporary difference can be utilized.

Deferred tax assets and liabilities are offset when there is a legally enforceable right to offset current tax assets and liabilities and when the deferred tax balances relate to the same taxation authority. Current tax assets and tax liabilities are offset when the Company has a legally enforceable right to offset and intends either to settle on a net basis, or to realize the assets and settle the liability simultaneously.

2.21 Employee Benefits

(a) Short-term employee benefits

Short-term employee benefits are employee benefits that are expected to be settled wholly before twelve months after the end of the annual reporting period in which the employees render the related services. When an employee has rendered service to the Company during an accounting period, the Company recognizes the undiscounted amount of short-term employee benefits expected to be paid in exchange for that service.

(b) Other long-term employee benefits

Other long-term employee benefits include employee benefits that are settled beyond 12 months after the end of the period in which the employees render the related service, and are calculated at the present value of the amount of future benefit that employees have earned in return for their service in the current and prior periods. Any changes from re-measurements are recognized through profit or loss in the period in which they arise.

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(c) Post-employment benefits: defined benefit plan

Generally, post-employment benefits are payable after the completion of employment, and the benefit amount depended on the employee's age, periods of service or salary levels. The liability recognized in the statement of financial position in respect of defined benefit pension plans is the present value of the defined benefit obligation at the end of the reporting period less the fair value of plan assets. The defined benefit obligation is calculated annually by independent actuaries using the projected unit credit method. The present value of the defined benefit obligation is determined by discounting the estimated future cash outflows using interest rates of high-quality corporate bonds that are denominated in the currency in which the benefits will be paid, and that have terms approximating to the terms of the related obligation. Remeasurement gains and losses arising from experience adjustments and changes in actuarial assumptions are recognized in the period in which they occur, directly in other comprehensive income.

Changes in the present value of the defined benefit obligation resulting from plan amendments or curtailments are recognized immediately in profit or loss as past service costs.

2.22 Contributed Capital

Ordinary shares are classified as equity. Transaction costs directly related to the issuance of ordinary shares and share options are recognized as a deduction from equity.

2.23 Earnings per Share

The Company presents basic and diluted earnings per share (EPS) data for its ordinary shares. Basic EPS is calculated by dividing the profit or loss attributable to ordinary shareholders of the Company by the weighted average number of ordinary shares outstanding during the period, adjusted for own shares held. The Company has no dilutive potential shares, therefore diluted loss per share is equal to the basic earnings (loss) per share.

2.24 Revenue Recognition

(a) Identification of performance obligation

- CMO business

The Company manufactures and supplies made-to-order bio-medicine. Services mandatorily rendered prior to a production of bio-medicine do not form a part of a performance obligation since they are considered as a set-up process. Only the sales of bio-medicine are recognized as performance obligations.

- CDO business

The Company provides cell line and upstream/downstream process development services. As each research and development stages is linked together and is highly interdependent and highly interrelated; therefore, the Company recognizes a contract as single performance obligation.

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(b) Performance obligation satisfied over time

- CMO business

Goods manufactured by the Company do not have an alternative use to the Company, and it has an enforceable right to payment for performance completed to date. This is because that the Company produces bio-medicine at a quality that a customer requires and is entitled to reimbursement of the costs incurred to date, including a reasonable margin if a customer terminates a contract. Therefore, the Company recognizes revenue and the associated costs over time as it satisfies its performance obligation over a contract period.

Revenue for bio-medicine other than payments related to transfer of technology and engineering batch production is recognized as revenues to the extent of the consideration equivalent to the Company's right to request for payment. Under the practical expedient of Korean IFRS 1115, the transaction price allocated to these unsatisfied contracts is not disclosed.

- CDO business

Revenue is recognized over time during a contract period as a customer simultaneously receives and consumes the benefits provided by the Company as the Company performs research and development services.

(c) Costs to fulfill a contract

If the costs incurred in fulfilling a contract with a customer are related directly to a contract or to an anticipated contract that the entity can specifically identify, generate or enhance resources of the entity that will be used in satisfying (or in continuing to satisfy) performance obligations in the future, and are expected to be recovered, the costs are recognized as assets. And the costs to fulfill a contract are recognized as cost of revenue based on percentage of completion.

(d) Contract assets and contract liabilities

- CMO business

The Company recognizes the amount of the advance consideration by customers related to transfer of technology to manufacture products that customer request and engineering batch production as contract liabilities and related costs as contract assets. Such activities are required in advance for production of goods ordered but do not transfer goods or services to a customer. Accordingly, they are not included in a performance obligation.

- CDO business

In a CDO business, a contract asset is an entity's right to receive consideration in exchange for goods or services that the entity has transferred to a customer, and a contract liability is the entity's obligation to transfer goods or services to a customer for which the entity has received consideration (or the amount is due) from the customer.

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(e) Variable consideration

If the consideration promised in a contract includes a variable amount, the Company estimates the amount of consideration to which the Company will be entitled in exchange for transferring the promised goods or services to a customer. An amount of consideration can vary because of discounts, refunds, credits, price concessions, penalties or other similar items. The promised consideration can also vary if an entity's entitlement to the consideration is contingent on the occurrence or non-occurrence of a future event.

(f) Recognition of receivable

A receivable is recognized when the control of bio-medicine has been transferred as this is the point in time that the consideration is unconditional because only the passage of time is required before the payment is due.

2.25 Leases

As explained in Note 2.2.1 above, the Company has changed its accounting policy for leases. The impact of the new accounting policies is disclosed in Note 2.2.1.

(a) Accounting policies of lessor

Lease income from operating leases where the Company is a lessor is recognized in income on a straight-line basis over the lease term. Initial direct costs incurred in obtaining an operating lease are added to the carrying amount of the underlying asset and recognized as expense over the lease term on the same basis as lease income. The respective leased assets are included in the statement of financial position based on their nature. The Company did not need to make any adjustments to the accounting for assets held as lessor as a result of adopting the new leasing standard.

(a) Accounting policies of lessee

The Company leases various offices, equipment and vehicles. Lease contracts are typically made for fixed periods of 2 to 3 years, but may have extension options.

Extension and termination options are included in a number of leases. These terms are used to maximize operational flexibility in terms of managing contracts. The majority of extension and termination options held are exercisable only by the Company and not by the respective lessor.

Lease terms are negotiated on an individual basis and contain a wide range of different terms and conditions. The lease agreements do not impose any covenants, but leased assets may not be used as security for borrowing purposes.

Until the 2018 financial year, leases of property, plant and equipment were classified as either finance or operating leases. Payments made under operating leases (net of any incentives received from the lessor) were charged to profit or loss on a straight-line basis over the period of the lease.

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From January 1, 2019, leases are recognized as a right-of-use asset and a corresponding liability at the date at which the leased asset is available for use by the Company.

Assets and liabilities arising from a lease are initially measured on a present value basis. Lease liability measurement also include payments to be made in option periods if the lessee is reasonably certain to exercise an option to extend the lease.

The lease payments are discounted using the interest rate implicit in the lease. If that rate cannot be determined, the lessee's incremental borrowing rate is used, being the rate that the lessee would have to pay to borrow the funds necessary to obtain an asset of similar value in a similar economic environment with similar terms and conditions.

The Company is exposed to potential future increases in variable lease payments based on an index or rate, which are not included in the lease liability until they take effect. When adjustments to lease payments based on an index or rate take effect, the lease liability is reassessed and adjusted against the right-of-use asset.

Each lease payment is allocated between the liability and finance cost. The finance cost is charged to profit or loss over the lease period so as to produce a constant periodic rate of interest on the remaining balance of the liability for each period.

Right-of-use assets are measured at cost comprising the following:

- the amount of the initial measurement of lease liability
- any lease payments made at or before the commencement date less any lease incentives received
- any initial direct costs, and
- restoration costs

The right-of-use asset is depreciated over the shorter of the asset's useful life and the lease term on a straight-line basis. If the Company is reasonably certain to exercise a purchase option, the right-of-use asset is depreciated over the underlying asset's useful life. The Company elected not to apply that revaluation model to buildings held by the Company that are presented in the right-of-use assets.

Short-term leases are leases with a lease term of 12 months or less. Low-value assets comprise IT-equipment and small items of office furniture. Payments associated with short-term leases and leases of low-value assets are recognized as an expense in profit or loss.

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2.26 Approval of Issuance of the Financial Statements

The financial statements 2019 were approved for issue by the Board of Directors on February 28, 2020 and are subject to change with the approval of shareholders at their Annual General Meeting.

3. Critical Accounting Estimates and Assumptions

The preparation of financial statements requires the Company to make estimates and assumptions concerning the future. Management also needs to exercise judgement in applying the Company's accounting policies. Estimates and assumptions are continually evaluated and are based on historical experience and other factors, including expectations of future events that are believed to be reasonable under the circumstances. As the resulting accounting estimates will, by definition, seldom equal the related actual results, it can contain a significant risk of causing a material adjustment.

The estimates and assumptions that have a significant risk of causing a material adjustment to the carrying amounts of assets and liabilities within the next financial year are discussed below. Additional information of significant judgement and assumptions of certain items are included in relevant notes.

(a) Income taxes

The Company's taxable income generated from these operations are subject to income taxes based on tax laws and interpretations of tax authorities in numerous jurisdictions. There are many transactions and calculations for which the ultimate tax determination is uncertain (Note 29).

If certain portion of the taxable income is not used for investments or increase in wages or dividends in accordance with the *Tax System for Recirculation of Corporate Income*, the Company is liable to pay additional income tax calculated based on the tax laws. The new tax system is effective for three years from 2015. Accordingly, the measurement of current and deferred income tax is affected by the tax effects from the new tax system. As the Company's income tax is dependent on the investments, increase in wages and dividends, there is an uncertainty measuring the final tax effects.

(b) Fair value of financial instruments

The fair value of financial instruments that are not traded in an active market is determined by using valuation techniques. The Company uses its judgment to select a variety of methods and make assumptions that are mainly based on market conditions existing at the end of each reporting period (Note 5).

(c) Impairment of financial assets

The provisions for impairment for financial assets are based on assumptions about risk of default and expected loss rates. The Company uses judgement in making these assumptions and selecting the inputs to the impairment calculation based on the Company's past history, existing

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market conditions as well as forward looking estimates at the end of each reporting period.

(d) Impairment of investments in associates and joint ventures

The Company reviews whether there are any impairment indicators of investments in associates and joint ventures annually at the end of each reporting period. An impairment assessment is performed when it is determined that there is an indicator of impairment or considering the material amount of investments at the end of every year. The carrying value is compared to the recoverable amount, which is the higher of value in use and the fair value less costs to sell. An impairment loss is recognized immediately as an expense and is subsequently reversed as recoverable amount of the investment subsequently increases.

(e) Impairment of intangible assets that have an indefinite useful life

The Company reviews whether there are any impairment indicators of intangible assets that have an indefinite useful life annually at the end of each reporting period. The recoverable amount of a cash generating unit (CGU) or assets is determined based on value-in-use calculations. These calculations require estimates.

(f) Net defined benefit liability

The present value of net defined benefit liability depends on a number of factors that are determined on an actuarial basis using a number of assumptions including the discount rate (Note 22).

(g) Impairment of costs to fulfill a contract

The Company reviews whether there are any impairment indicators of costs to fulfil a contract at the end of each reporting period. The Company recognizes the impairment by comparing the revenue expected to receive for goods or service related to the costs to fulfil a contract with expected cost that are directly attributable. These calculations require estimates.

(g) Lease

In determining the lease term, management considers all facts and circumstances that create an economic incentive to exercise an extension option, or not exercise a termination option. Extension options (or periods after termination options) are only included in the lease term if the lease is reasonably certain to be extended (or not terminated).

The lease term is reassessed if an option is actually exercised (or not exercised) or the Company becomes obliged to exercise (or not exercise) it. The assessment of reasonable certainty is only revised if a significant event or a significant change in circumstances occurs, which affects this assessment, and that is within the control of the lessee.

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4. Operating Segments

The Company has disclosed operating segments in the financial statements in accordance with Korean IFRS 1108. The Company has one reportable segment, 'Biosimilar manufacturing', as at December 31, 2019.

Details of customers, who take over 10% of the Company's total revenue for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Client A	₩ 188,387,885	₩ 99,980,807
Client B	183,125,111	225,444,041
Client C	107,068,561	109,530,876
Client D	82,980,005	43,336,414
	<u>₩ 561,561,562</u>	<u>₩ 478,292,138</u>

Geographic revenue for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Domestic	₩ 207,119,950	₩ 227,949,588
Europe	418,234,272	290,009,678
USA	62,384,398	10,566,375
Other	13,853,240	7,280,160
	<u>₩ 701,591,860</u>	<u>₩ 535,805,801</u>

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5. Financial Instruments

(a) Financial assets

Categories of financial assets as at December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)

(in thousands of Korean won)

	2019					
	Financial assets at fair value through profit or loss		Financial assets measured at amortized cost		Total	
Current assets						
Cash and cash equivalents	₩	-	₩	125,100,653	₩	125,100,653
Short-term financial instruments		-		320,000,000		320,000,000
Trade and other receivables		-		187,118,116		187,118,116
Loans		-		-		-
Deposits provided		-		34,782,011		34,782,011
Financial assets at fair value through profit or loss		210,040,293		-		210,040,293
		210,040,293		667,000,780		877,041,073
Non-current assets						
Long-term financial instruments		-		7,000		7,000
Loans		-		400,000		400,000
Deposits provided		-		8,095,962		8,095,962
Derivative assets ¹		1,002,990		-		1,002,990
		1,002,990		8,502,962		9,505,952
	₩	211,043,283	₩	675,503,742	₩	886,547,025

(in thousands of Korean won)

(in thousands of Korean won)

	2018					
	Financial assets at fair value through profit or loss		Financial assets measured at amortized cost		Total	
Current assets						
Cash and cash equivalents	₩	-	₩	17,621,905	₩	17,621,905
Short-term financial instruments		-		550,000,000		550,000,000
Trade and other receivables		-		121,733,350		121,733,350
Loans		-		600,000		600,000
Deposits provided		-		20,279,609		20,279,609
Financial assets at fair value through profit or loss		540,080,517		-		540,080,517
		540,080,517		710,234,864		1,250,315,381
Non-current assets						
Long-term financial instruments		-		7,000		7,000
Loans		-		-		-
Deposits provided		-		24,230,355		24,230,355
Derivative assets ¹		-		-		-
		-		24,237,355		24,237,355
	₩	540,080,517	₩	734,472,219	₩	1,274,552,736

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¹ Details of the derivative from currency interest swap into which the Company entered for foreign currency borrowings as at December 31, 2019, are as follows:

Classification	Category	Financial institution	Contracted amount	Contract date	Maturity date	Currency(*)		Contracted exchange rate	Interest rate(%)	
						Sell	Buy		Receiving	Paying
Trading	Currency interest swap	KEB Hana Bank	USD 30,000,000	2018-05-31	2023-05-31	USD	KRW	1,080.00	Libor(3M) + 1.70	3.70

(*) It is stated based on the contract position at the point of borrowing.

(b) Financial liabilities

Categories of financial liabilities as at December 31, 2019 and 2018, are as follows:

		2019			
		Financial liabilities at fair value through profit or loss	Financial liabilities measured at amortized cost	Other liabilities	Total
Current liabilities					
Trade and other payables ¹	₩	-	₩ 68,251,491	₩ -	₩ 68,251,491
Lease liabilities		-	-	3,248,590	3,248,590
Debentures		-	-	-	-
Short-term borrowings		-	150,000,000	-	150,000,000
Current portion of long-term borrowings		-	251,650,000	-	251,650,000
		-	469,901,491	3,248,590	473,150,081
Non-current liabilities					
Lease liabilities		-	-	12,640,694	12,640,694
Debentures		-	189,755,125	-	189,755,125
Long-term borrowings		-	46,412,000	-	46,412,000
Derivative liabilities		-	-	-	-
		-	236,167,125	12,640,694	248,807,819
	₩	-	₩ 706,068,616	₩ 15,889,284	₩ 721,957,900

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		2018			
		Financial liabilities at fair value through profit or loss		Financial liabilities measured at amortized cost	Total
Current liabilities					
Trade and other payables ¹	₩	-	₩	144,927,592	₩ 144,927,592
Lease liabilities		-		-	-
Debentures		-		50,000,000	50,000,000
Short-term borrowings		-		90,000,000	90,000,000
Current portion of long- term borrowings		-		246,583,355	246,583,355
		-		531,510,947	531,510,947
Non-current liabilities					
Lease liabilities		-		-	-
Debentures		-		189,638,752	189,638,752
Long-term borrowings		-		296,474,000	296,474,000
Derivative liabilities		190,330		-	190,330
		190,330		486,112,752	486,303,082
	₩	190,330	₩	1,017,623,699	₩ 1,017,814,029

¹ The amounts exclude the accrued compensation for annual leaves, performance bonuses, and others payable to employees.

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(c) The fair values of financial instruments

The fair values of financial instruments, together with the carrying amounts shown in the statement of financial position as at December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)

	2019		2018	
	Carrying amount	Fair value	Carrying amount	Fair value
Assets measured at fair value				
Financial assets at fair value through profit or loss	₩ 210,040,293	₩ 210,040,293	₩ 540,080,517	₩ 540,080,517
Derivative assets	1,002,990	1,002,990	-	-
	<u>211,043,283</u>	<u>211,043,283</u>	<u>540,080,517</u>	<u>540,080,517</u>
Assets not measured fair value				
Cash and cash equivalents	125,100,653	125,100,653	17,621,905	17,621,905
Short-term financial instruments	320,000,000	320,000,000	550,000,000	550,000,000
Long-term financial instruments	7,000	7,000	7,000	7,000
Trade and other receivables	187,118,116	187,118,116	121,733,350	121,733,350
Loans	400,000	400,000	600,000	600,000
Deposits provided	42,877,973	42,877,973	44,509,965	44,509,965
	<u>675,503,742</u>	<u>675,503,742</u>	<u>734,472,220</u>	<u>734,472,220</u>
	<u>₩ 886,547,025</u>	<u>₩ 886,547,025</u>	<u>₩ 1,274,552,737</u>	<u>₩ 1,274,552,737</u>
Liabilities carried at fair value				
Derivative liabilities	₩ -	₩ -	₩ 190,330	₩ 190,330
Liabilities carried at amortized cost				
Trade and other payables	68,251,491	68,251,491	144,927,592	144,927,592
Debentures	189,755,125	189,755,125	239,638,752	239,638,752
Borrowings	448,062,000	448,062,000	633,057,355	633,057,355
	<u>706,068,616</u>	<u>706,068,616</u>	<u>1,017,623,699</u>	<u>1,017,623,699</u>
	<u>₩ 706,068,616</u>	<u>₩ 706,068,616</u>	<u>₩ 1,017,814,029</u>	<u>₩ 1,017,814,029</u>

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(d) Fair value hierarchy

The following presents financial instruments measured and recognized at fair value as at December 31, 2019 and 2018:

<i>(in thousands of Korean won)</i>		2019			
		Level 1	Level 2	Level 3	Total
Financial assets					
Financial assets at fair value through profit or loss	₩	-	₩ 210,040,293	₩ -	₩ 210,040,293
Derivative assets		-	1,002,990	-	1,002,990

<i>(in thousands of Korean won)</i>		2018			
		Level 1	Level 2	Level 3	Total
Financial assets					
Financial assets at fair value through profit or loss	₩	-	₩ 540,080,517	₩ -	₩ 540,080,517
Financial liabilities					
Derivative liabilities		-	190,330	-	190,330

(e) Valuation techniques and inputs

Valuation techniques and significant inputs used in measuring the fair value of financial instruments classified as level 2 and 3 as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	Fair value		Valuation techniques	Significant unobservable inputs
	2019	2018		
Financial assets at fair value through profit or loss	₩ 210,040,293	₩ 540,080,517	Market approach	-
Derivative assets	1,002,990	-	Discount cash flow model	Currency forward rate, discount rate, and others
Derivative liabilities	-	190,330		

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(f) Net Gains or Losses by Category of Financial Instruments

Net gains or losses on each category of financial instruments for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	Interest income (expenses)		Gain (loss) on valuation		Other income (expenses)¹		Net income (loss)	
	2019	2018	2019	2018	2019	2018	2019	2018
Financial assets at amortized cost	₩ 17,246,464	₩ 9,651,186	₩ -	₩ -	₩ (6,877,661)	₩ 1,500,490	₩ 10,368,803	₩ 11,151,676
Financial assets at fair value through profit or loss	-	-	45,759	80,517	361,459	301,927	407,218	382,444
Derivative assets	-	-	1,193,321	-	-	-	1,193,321	-
Financial liabilities at amortized cost	(25,614,722)	(15,838,589)	-	-	2,503,481	(521,627)	(23,111,241)	(16,360,216)
Financial liabilities at fair value through profit or loss	-	-	-	(190,330)	-	-	-	(190,330)
	<u>₩ (8,368,258)</u>	<u>₩ (6,187,403)</u>	<u>₩ 1,239,080</u>	<u>₩ (109,813)</u>	<u>₩ (4,012,721)</u>	<u>₩ 1,280,790</u>	<u>₩ (11,141,899)</u>	<u>₩ (5,016,426)</u>

¹ Other income and expenses include gain (loss) on foreign currency translations and transactions, and gain (loss) on disposal of financial assets at fair value through profit or loss.

6. Cash and Cash Equivalents

Cash and cash equivalents as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Demand deposits	₩ 125,100,653	₩ 17,621,905

7. Restricted Financial Instruments

Restriction of financial instruments as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018	Note
Long-term financial instruments	₩ 7,000	₩ 7,000	Deposits on checking accounts

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8. Trade and Other Receivables

Details of trade and other receivables as at December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)

	2019					
	Related parties		Others		Total	
Trade receivables	₩	98,332,044	₩	82,791,067	₩	181,123,111
Other receivables		187,456		1,869,989		2,057,445
Accrued income		753,409		3,184,150		3,937,559
	₩	99,272,909	₩	87,845,206	₩	187,118,115

(in thousands of Korean won)

	2018					
	Related parties		Others		Total	
Trade receivables	₩	65,408,494	₩	47,147,376	₩	112,555,870
Other receivables		121,565		5,159,933		5,281,498
Accrued income		799,878		3,096,104		3,895,982
	₩	66,329,937	₩	55,403,413	₩	121,733,350

The aging analysis of non-impaired trade and other receivables as at December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)

	2019		2018	
Receivables not past due	₩	171,651,512	₩	115,871,507
Past due				
Less than 1 month		15,315,193		5,861,843
Between 1 and 3 months		151,410		-
	₩	187,118,115	₩	121,733,350

The Company calculated an estimate of expected credit losses based on actual credit loss experience over the past two years and determined that it is not necessary to recognize allowance for trade and other receivables.

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9. Contract Assets and Liabilities

Changes in contract assets and liabilities arising from contracts with customers for year ended December 31, 2019, are as follows:

<i>(in thousands of Korean won)</i>	Beginning balance	Increase	Decrease	Ending balance
Contract assets - costs to fulfil a contract	₩ 95,743,870	₩ 70,960,964	₩ (16,857,247)	₩ 149,847,587
Contract assets - unbilled	834,768	7,672,591	(6,872,059)	1,635,300
Contract liabilities	101,104,790	93,011,436	(48,986,970)	145,129,256

Costs incurred in fulfilling a contract include costs for technology transfer and trial batches production in order to manufacture products that customers request. Such activities do not include transferring goods or services to a customer but are prerequisite for production of goods ordered. Accordingly, they do not consist as part of a performance obligation but are recognized as contract assets. Unbilled contract assets were recognized due to a difference between the extent of the Company's satisfaction of its performance obligation satisfied over time and the consideration billed to the customer. The Company recognizes any advance consideration received from a customer in relation to technology transfer and engineering batch production as contract liabilities.

Reversal of impairment loss of ₩ 2,263,141 thousand on contract assets is recognized during the year ended December 31, 2019.

Besides the above contract liabilities recognized, the Company classifies the amount received prior to transferring goods or services to the customer as 'advance receipts'. As at December 31, 2019, the balance of advance receipts is ₩ 147,373,367 thousand.

10. Inventories

Details of inventories as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019		
	Acquisition cost	Provision for loss on valuation	Book amount
Finished goods	₩ 129,653,310	₩ (7,710,652)	₩ 121,942,658
Work-in-process	141,173,973	(5,372,286)	135,801,687
Raw materials	111,630,168	-	111,630,168
Supplies	83,095,414	-	83,095,414
	<u>₩ 465,552,865</u>	<u>₩ (13,082,938)</u>	<u>₩ 452,469,927</u>

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(in thousands of Korean won)

	2018		
	Acquisition cost	Provision for loss on valuation	Book amount
Finished goods	₩ 84,434,000	₩ (6,442,855)	₩ 77,991,145
Work-in-process	77,594,002	(424,700)	77,169,302
Raw materials	83,483,857	-	83,483,857
Supplies	63,716,411	-	63,716,411
	<u>₩ 309,228,270</u>	<u>₩ (6,867,555)</u>	<u>₩ 302,360,715</u>

During the year ended December 31, 2019, loss on valuation of ₩ 6,215,383 thousand was recognized while reversal of loss on valuation of ₩ 1,932,843 thousand was recognized during the year ended December 31, 2018. The amounts are included in cost of revenue.

11. Other Financial Assets

Details of other financial assets as at December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)

	2019		2018	
	Current	Non-current	Current	Non-current
Deposits provided	₩ 34,782,011	₩ 8,095,962	₩ 20,279,609	₩ 24,230,355
Loans	-	400,000	600,000	-
Derivative assets	-	1,002,990	-	-
	<u>₩ 34,782,011</u>	<u>₩ 9,498,952</u>	<u>₩ 20,879,609</u>	<u>₩ 24,230,355</u>

12. Other Assets

Details of other assets as at December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)

	2019		2018	
	Current	Non-current	Current	Non-current
Advance payments	₩ 18,390	₩ -	₩ -	₩ -
Prepaid expenses	5,919,923	956,721	5,508,443	1,902,973
	<u>₩ 5,938,313</u>	<u>₩ 956,721</u>	<u>₩ 5,508,443</u>	<u>₩ 1,902,973</u>

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13. Investments in Associates and Joint Ventures

Investments in associates and joint ventures as at December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)	Major business location	Primary business	Fiscal date	2019		2018	
				Book amount	Percentage of ownership	Book amount	Percentage of ownership
Archigen Biotech Limited	The United Kingdom	Bio-medicine business	2019.12.31	₩ 2,765,976	50.00%	₩ -	50.00%
Samsung Bioepis Co., Ltd. ¹	Republic of Korea	Biosimilar research and commercialization	2019.12.31	2,649,965,646	50.00%	2,561,802,521	50.00%
				<u>₩2,652,731,622</u>		<u>₩2,561,802,521</u>	

¹ Samsung Bioepis Co., Ltd. was established in 2012 in a joint venture between the Company and Biogen Therapeutics Inc. for development and commercialization of biosimilar. The Company had classified Samsung Bioepis Co., Ltd. as a subsidiary until 2015 when the Company lost its control that potential voting rights held by Biogen Therapeutics Inc. were changed to substantive rights to Samsung Bioepis Co., Ltd. and related activities could not be controlled unilaterally due to the composition of the board of directors and the quorum for voting at shareholder meeting under the agreement. According to the joint venture agreement between the Company and Biogen Therapeutics Inc., Samsung Bioepis Co., Ltd. entered into a contract for intellectual property and others with Biogen Therapeutics Inc. and other companies

Biogen Therapeutics Inc. exercised the call option to purchase shares of Samsung Bioepis Co., Ltd., on June 29, 2018. All legal procedures related to exercising the call option were completed. Accordingly, as at December 31, 2019, the percentage of the Company's ownership is 50%+1 share.

Changes in investments in associates and joint ventures for the years ended December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)	2019				
	Beginning balance	Acquisition	Share of profit (loss)	Other changes ¹	Ending balance
Archigen Biotech Limited	₩ -	₩ 18,101,000	₩ (15,144,614)	₩ (190,410)	₩ 2,765,976
Samsung Bioepis Co., Ltd.	<u>2,561,802,521</u>	<u>-</u>	<u>88,084,729</u>	<u>78,396</u>	<u>2,649,965,646</u>
	<u>₩ 2,561,802,521</u>	<u>₩ 18,101,000</u>	<u>₩ 72,940,115</u>	<u>₩ (112,014)</u>	<u>₩ 2,652,731,622</u>

¹ Other changes include changes in equity of investee and changes in exchange rate.

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(in thousands of Korean won)	2018						
	Beginning balance	Application of Korean IFRS 1115	Acquisition	Share of loss	Other changes ¹	Disposal	Ending balance
Archigen Biotech Limited ²	₩ 10,060,921	₩ -	₩ 16,405,000	₩ (27,852,433)	₩ 1,386,512	₩ -	₩ -
Samsung Bioepis Co., Ltd.	4,998,173,084	(34,300,467)	-	(101,484,900)	(746,384)	(2,299,838,812)	2,561,802,521
	<u>₩5,008,234,005</u>	<u>₩ (34,300,467)</u>	<u>₩ 16,405,000</u>	<u>₩ (129,337,333)</u>	<u>₩ 640,128</u>	<u>₩ (2,299,838,812)</u>	<u>₩ 2,561,802,521</u>

¹ Other changes include changes in equity of investee and changes in exchange rate.

² As the carrying amount became less than zero during the year ended December 31, 2018, the application of the equity method was discontinued. Accordingly, share of loss which was not recognized during the year ended December 31, 2018, is ₩ 8,328,897 thousand, and the amount was recognized during the year ended December 31, 2019.

Financial information of investees as at December 31, 2019 and 2018 and for the years ended December 31, 2019 and 2018, are summarized as follows:

(in thousands of Korean won)	2019						
	Current assets	Non-current assets	Current liabilities	Non-current liabilities	Revenue	Profit (loss) for the year	Total comprehen- sive income (loss)
Archigen Biotech Limited	₩ 17,872,041	₩ 1,937	₩ 12,342,027	₩ -	₩ -	₩(14,811,652)	₩(14,446,884)
Samsung Bioepis Co., Ltd. ¹	1,677,876,662	824,352,556	1,611,430,379	224,917,308	765,859,497	263,418,473	263,747,201

¹ The amounts are based on financial information of Samsung Bioepis Co., Ltd. before adjustment of fair value of net assets.

(in thousands of Korean won)	2018						
	Current assets	Non-current assets	Current liabilities	Non-current liabilities	Revenue	Loss for the year	Total comprehen- sive loss
Archigen Biotech Limited	₩ 11,705,529	₩ 176,477	₩ 28,512,961	₩ 26,840	₩ -	₩ (95,440,595)	₩ (95,440,595)
Samsung Bioepis Co., Ltd. ¹	1,332,991,753	564,198,972	1,229,279,243	265,777,152	368,733,526	(54,956,629)	(56,585,151)

¹ The amounts are based on financial information of Samsung Bioepis Co., Ltd. before adjustment of fair value of net assets.

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Reconciliation of the investees' financial information to the carrying value of the Company's interest in associates and joint ventures as at December 31, 2019 and 2018, are as follows:

(in thousands of
Korean won)

	2019					
	Net assets	Percentage of ownership	The Company's share of net assets	Investment difference ¹	Unrealized intercompany transactions	Book amount
Archigen Biotech Limited	₩ 5,531,951	50.00%	₩ 2,765,976	₩ -	₩ -	₩ 2,765,976
Samsung Bioepis Co., Ltd.	665,881,531	50.00%	332,940,798	2,383,358,064	(66,333,216)	2,649,965,646

¹ Investment difference includes adjustment of net assets value at the point of loss of control and a tentative amount related to irregular capital increase, and the amortization effect.

(in thousands of
Korean won)

Korean won)	2018						
	Net assets	Percentage of ownership	The Company's share of net assets	Investment difference ¹	Unrealized intercompany transactions	Unrecognized loss	Book amount
Archigen Biotech Limited	₩(16,657,794)	50.00%	₩ (8,328,897)	₩ -	₩ -	₩ 8,328,897	₩ -
Samsung Bioepis Co., Ltd.	402,134,330	50.00%	201,170,693	2,401,335,219	(40,703,391)	-	2,561,802,521

¹ Investment difference includes adjustment of net assets value at the point of loss of control, the effect of selective capital increase, and the amortization effect.

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14. Investment Properties

Details of investment properties as at December 31, 2019, are as follows:

(in thousands of Korean won)

	2019		
	Acquisition cost	Accumulated depreciation	Book amount
Buildings	₩ 20,480,236	₩ (341,337)	₩ 20,138,899
Structures	45,052	(751)	44,301
Others	577,893	-	577,893
	<u>₩ 21,103,181</u>	<u>₩ (342,088)</u>	<u>₩ 20,761,093</u>

Changes in investment properties for the year ended December 31, 2019, are as follows:

(in thousands of Korean won)

	2019					
	Beginning balance	Acquisition	Disposal and others	Depreciation	Others	Ending balance
Buildings	₩ -	₩ 20,480,236	₩ -	₩ (341,337)	₩ -	₩ 20,138,899
Structures	-	45,052	-	(751)	-	44,301
Others	-	577,893	-	-	-	577,893
	<u>₩ -</u>	<u>₩ 21,103,181</u>	<u>₩ -</u>	<u>₩ (342,088)</u>	<u>₩ -</u>	<u>₩ 20,761,093</u>

Depreciation of investment properties was recognized as selling, general and administrative expenses.

Fair value of investment properties as at December 31, 2019, is ₩ 20,163 million. The valuation of investment properties is performed by an independent appraiser.

As at December 31, 2019, investment properties consist of buildings and structures leased to Samsung Bioepis Co., Ltd., as an associate of the Company. During the year ended December 31, 2019, rental income from investment property is ₩ 875 million.

The future minimum lease payments expected to be received in relation to the above operating lease agreement for investment property as at December 31, 2019, are as follows:

(in thousands of Korean won)

	2019
Within one year	₩ 1,500,000
Later than one year but not later than two years	250,000
	<u>₩ 1,750,000</u>

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15. Property, Plant and Equipment

Property, plant and equipment as at December 31, 2019 and 2018, consist of:

(in thousands of Korean won)	2019			2018		
	Acquisition cost	Accumulated depreciation	Book amount	Acquisition cost	Accumulated depreciation	Book amount
Buildings	₩ 780,731,716	₩ (61,738,343)	₩ 718,993,373	₩ 767,851,048	₩ (42,227,547)	₩ 725,623,501
Structures	10,341,793	(1,154,750)	9,187,043	7,731,647	(660,340)	7,071,307
Machinery	1,207,232,972	(288,333,414)	918,899,558	1,100,464,489	(195,747,495)	904,716,994
Vehicles	1,007,337	(744,034)	263,303	959,580	(582,548)	377,032
Others	43,725,422	(24,990,728)	18,734,694	39,826,906	(17,010,526)	22,816,380
Construction-in-progress	36,714,553	-	36,714,553	57,115,774	-	57,115,774
	<u>₩ 2,079,753,793</u>	<u>₩ (376,961,269)</u>	<u>₩ 1,702,792,524</u>	<u>₩ 1,973,949,444</u>	<u>₩ (256,228,456)</u>	<u>₩ 1,717,720,988</u>

Changes in property, plant and equipment for the years ended December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)	2019					
	Beginning balance	Acquisition	Disposal and others	Depreciation	Others	Ending balance
Buildings	₩ 725,623,501	₩ 130,657	₩ (149,418)	₩ (19,523,464)	₩ 12,912,097	₩ 718,993,373
Structures	7,071,307	-	-	(222,519)	2,338,255	9,187,043
Machinery	904,716,994	1,316,151	(465,363)	(93,109,928)	106,441,704	918,899,558
Vehicles	377,032	47,757	-	(161,486)	-	263,303
Others	22,816,380	2,325,435	(36,187)	(8,874,953)	2,504,019	18,734,694
Construction-in-progress	57,115,774	104,849,634	-	-	(125,250,855)	36,714,553
	<u>₩ 1,717,720,988</u>	<u>₩ 108,669,634</u>	<u>₩ (650,968)</u>	<u>₩ (121,892,350)</u>	<u>₩ (1,054,780)</u>	<u>₩ 1,702,792,524</u>

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(in thousands of Korean won)	2018					
	Beginning balance	Acquisition	Disposal and others	Depreciation	Others	Ending balance
Buildings	₩ 589,398,424	₩ 7,373,168	₩ -	₩ (15,856,564)	₩ 144,708,473	₩ 725,623,501
Structures	7,264,598	-	-	(193,291)	-	7,071,307
Machinery	596,362,685	3,224,504	(608,528)	(63,364,286)	369,102,619	904,716,994
Vehicles	486,816	44,500	-	(154,284)	-	377,032
Others	10,975,533	4,407,724	(6,368)	(5,953,937)	13,393,428	22,816,380
Construction-in-progress	308,844,562	275,484,587	-	-	(527,213,375)	57,115,774
	<u>₩ 1,513,332,618</u>	<u>₩ 290,534,483</u>	<u>₩ (614,896)</u>	<u>₩ (85,522,362)</u>	<u>₩ (8,855)</u>	<u>₩ 1,717,720,988</u>

Classification of depreciation for the years ended December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)	2019	2018
Cost of revenue	₩ 116,179,700	₩ 73,730,682
Selling, general and administrative expenses	5,712,650	11,791,680
	<u>₩ 121,892,350</u>	<u>₩ 85,522,362</u>

In 2019, there are no capitalization of borrowing costs. ₩ 8,911,242 thousand of borrowing costs were capitalized as construction-in-progress for the year ended December 31, 2018. Capitalized interest rate for the general borrowings used to calculate the borrowing costs capitalization for the year ended December 31, 2018, is 3.45%.

16. Intangible Assets

Intangible assets as at December 31, 2019 and 2018, consist of:

(in thousands of Korean won)	2019			
	Acquisition cost	Accumulated amortization and impairment	Government grants	Book amount
Software	₩ 40,996,127	₩ (24,968,352)	₩ -	₩ 16,027,775
Others	1,012,652	(880,002)	-	132,650
Memberships	4,243,388	(521,756)	-	3,721,632
Land use rights	69,525,088	(12,051,015)	(57,474,073)	-
Construction-in-progress	1,945,458	-	-	1,945,458
	<u>₩ 117,722,713</u>	<u>₩ (38,421,125)</u>	<u>₩ (57,474,073)</u>	<u>₩ 21,827,515</u>

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(in thousands of Korean won)

	2018			
	Acquisition cost	Accumulated amortization and impairment	Government grants	Book amount
Software	₩ 35,350,378	₩ (20,503,970)	₩ -	₩ 14,846,408
Others	1,059,446	(816,790)	-	242,656
Memberships	2,573,188	(10,756)	-	2,562,432
Land use rights	69,525,088	(10,660,513)	(58,864,575)	-
Construction-in-progress	2,330,138	-	-	2,330,138
	<u>₩ 110,838,238</u>	<u>₩ (31,992,029)</u>	<u>₩ (58,864,575)</u>	<u>₩ 19,981,634</u>

Changes in intangible assets for the years ended December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)

	2019						
	Beginning balance	Acquisition	Disposal and others	Amortization	Amortization of government grants	Others	Ending balance
Software	₩ 14,846,408	₩ 1,242,021	₩ (178,850)	₩ (6,200,796)	₩ -	₩ 6,318,992	₩ 16,027,775
Others	242,656	-	(18,717)	(91,289)	-	-	132,650
Memberships	2,562,432	1,670,200	(511,000)	-	-	-	3,721,632
Land use rights	-	-	-	(1,390,502)	1,390,502	-	-
Construction-in-progress	2,330,138	5,033,399	-	-	-	(5,418,079)	1,945,458
	<u>₩ 19,981,634</u>	<u>₩ 7,945,620</u>	<u>₩ (708,567)</u>	<u>₩ (7,682,587)</u>	<u>₩ 1,390,502</u>	<u>₩ 900,913</u>	<u>₩ 21,827,515</u>

(in thousands of Korean won)

	2018						
	Beginning balance	Acquisition	Disposal and others	Amortization	Amortization of government grants	Others	Ending balance
Software	₩ 12,502,855	₩ 2,087,345	₩ -	₩ (5,761,652)	₩ -	₩ 6,017,860	₩ 14,846,408
Others	409,197	-	-	(166,541)	-	-	242,656
Memberships	2,562,188	11,000	(10,756)	-	-	-	2,562,432
Land use rights	-	-	-	(1,390,502)	1,390,502	-	-
Construction-in-progress	3,524,763	4,814,380	-	-	-	(6,009,005)	2,330,138
	<u>₩ 18,999,003</u>	<u>₩ 6,912,725</u>	<u>₩ (10,756)</u>	<u>₩ (7,318,695)</u>	<u>₩ 1,390,502</u>	<u>₩ 8,855</u>	<u>₩ 19,981,634</u>

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Classification of amortization for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019		2018	
Cost of revenue	₩	2,752,960	₩	3,104,489
Selling, general and administrative expenses		3,539,125		2,823,704
	₩	6,292,085	₩	5,928,193

The Company performed an impairment test for intangible assets with indefinite useful lives and recognized an impairment loss of ₩ 359,600 thousand and ₩ 10,756 thousand for the years ended December 31, 2019 and 2018, respectively. The recoverable amount of an asset or CGU is the greater of its value in use and its fair value less costs to sell. The value in use is estimated by applying a discount rate that reflect current market assessments of the time value of money and the risks specific to the asset or CGU for which estimated future cash flows have not been adjusted, to the estimated future cash flows expected to be generated by the asset or CGU.

17. Debenture and Borrowings

Debentures and borrowings as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019		2018	
	Current	Non-current	Current	Non-current
Debentures	₩ -	₩ 189,755,125	₩ 50,000,000	₩ 189,638,752
Short-term borrowings	150,000,000	-	90,000,000	-
Long-term borrowings	251,650,000	46,412,000	246,583,355	296,474,000
	₩ 401,650,000	₩ 236,167,125	₩ 386,583,355	₩ 486,112,752

Details of debentures as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	Issuance date	Maturity date	Interest rate (%)	2019	2018
Long-term debentures:					
The 1st placement debenture (non-guaranteed)	2014-10-24	2019-10-24	3.48	₩ -	₩ 50,000,000
The 6-1st placement debenture(non-guaranteed)	2018-04-27	2021-04-27	3.10	150,000,000	150,000,000
The 6-2nd placement debenture(non-guaranteed)	2018-04-27	2023-04-27	3.53	40,000,000	40,000,000
Discount on debentures				(244,875)	(361,248)
Current portion				-	(50,000,000)
				₩ 189,755,125	₩ 189,638,752

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Details of short-term borrowings as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	Description	Maturity date	Interest rate (%)	2019	2018
Korea Development Bank	Loans for equipment	2019-09-28	CD(91D)+0.90	₩ -	₩ 40,000,000
Mizuho Bank	Loans for equipment	2019-12-20	CD(91D)+0.60	-	50,000,000
KEB Hana Bank	Loans for equipment	2020-11-13	CD(91D)+1.35	50,000,000	-
Mizuho Bank	Loans for equipment	2020-12-18	CD(91D)+0.46	70,000,000	-
Sumitomo Mitsui Bank	Loans for operation	2020-01-31	CD(91D)+0.55	30,000,000	-
				<u>₩ 150,000,000</u>	<u>₩ 90,000,000</u>

Details of long-term borrowings as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	Description	Maturity date	Interest rate (%)	2019	2018
Korea Development Bank	Loans for equipment	2019-07-24	3.24 ~ 3.57	₩ -	₩ 50,000,000
		2020-06-12 ¹	2.86 ~ 2.90	12,500,000	37,500,000
		2020-12-15 ¹	2.86	20,000,000	40,000,000
Export-Import Bank of Korea	Overseas investment loans	2019-06-15	LIBOR+1.50	-	15,094,350
Kookmin Bank	Loans for equipment	2020-08-11 ¹	CD(91D)+1.54	18,750,000	43,750,000
		2020-09-28	CD(91D)+1.43	60,000,000	60,000,000
KEB Hana Bank	Loans for equipment	2019-11-14	CD(91D)+1.45	-	50,000,000
		2021-02-26 ¹	CD(91D)+1.60	500,000	800,000
		2023-05-31	LIBOR+1.70	34,734,000	33,543,000
NongHyup Bank	Loans for equipment	2019-11-25	MOR(1Y)+1.52	-	50,000,000
Woori Bank	Loans for equipment	2020-09-28	CD(91D)+1.43	70,000,000	70,000,000
Shinhan Bank	Loans for equipment	2020-09-28	CD(91D)+1.44	70,000,000	70,000,000
Industrial and Commercial Bank of China, Ltd.	Loans for equipment in foreign currency	2019-06-29	LIBOR+1.50	-	11,189,005
Sumitomo Mitsui Bank	Overseas investment loans	2021-03-22	LIBOR+1.10	11,578,000	11,181,000
Current portion				<u>(251,650,000)</u>	<u>(246,583,355)</u>
				<u>₩ 46,412,000</u>	<u>₩ 296,474,000</u>

¹ The borrowings have the condition for instalment payment every three months starting from two years before their maturities.

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The repayment schedule of debentures and borrowings as at December 31, 2019, are as follows:

<i>(in thousands of Korean won)</i>	Debentures	Short-term borrowings	Long-term borrowings	Total
2020.01.01 ~ 2020.12.31	₩ -	₩ 150,000,000	₩ 251,650,000	₩ 401,650,000
2021.01.01 ~ 2021.12.31	150,000,000	-	11,678,000	161,678,000
2022.01.01 ~ 2022.12.31	-	-	-	-
2023.01.01 ~	40,000,000	-	34,734,000	74,734,000
	<u>₩ 190,000,000</u>	<u>₩ 150,000,000</u>	<u>₩ 298,062,000</u>	<u>₩ 638,062,000</u>

18. Trade and Other Payables

Details of trade and other payables as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019		2018	
	Current	Non-current	Current	Non-current
Trade payables	₩ 12,289,708	₩ -	₩ 3,810,760	₩ -
Other payables	44,961,882	-	116,751,027	-
Accrued expenses	43,876,471	8,105,041	49,776,507	3,988,729
Total ¹	<u>₩ 101,128,061</u>	<u>₩ 8,105,041</u>	<u>₩ 170,338,294</u>	<u>₩ 3,988,729</u>

¹ Trade and other payables to related parties as at December 31, 2019 and 2018, are ₩ 9,492,306 thousand and ₩ 82,709,228 thousand, respectively (Note 32).

19. Emission Liability

(a) Emissions rights

The Company has been participating in Emission Trading Scheme in accordance with the Act on Allocation and Trading of Greenhouse Gas Emission since 2017. For 2019, the Company has been granted free of charge emission rights of 61,444 tons of greenhouse gas emissions for its first and second plants, and the Company plans to apply for additional allocation after confirming the emissions during 2019 from its new facility, third plant. There are no emission rights provided as collateral as at December 31, 2019.

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(b) Emission liability

Changes in emission liability for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019			
	Beginning balance	Increase	Decrease	Ending balance
Emission liability	₩ 1,493,982	₩ 553,853	₩ (1,493,982)	₩ 553,853

<i>(in thousands of Korean won)</i>	2018			
	Beginning balance	Increase	Decrease	Ending balance
Emission liability	₩ 502,080	₩ 1,493,982	₩ (502,080)	₩ 1,493,982

The Company estimated that it emitted 119,296 tons of carbon dioxide during the year ended December 31, 2019.

20. Lease

The amounts relating to leases recognized as at December 31, 2019 on the statement of financial position, are as follows:

<i>(in thousands of Korean won)</i>	December 31, 2019	January 1, 2019
Right-of-use assets¹		
Buildings	₩ 15,710,296	₩ 18,492,003
Machinery	241,533	15,919
Vehicles	167,054	212,103
	<u>₩ 16,118,883</u>	<u>₩ 18,720,025</u>
Lease liabilities		
Current	₩ 3,248,590	₩ 3,332,778
Non-current	12,640,694	14,444,560
	<u>₩ 15,889,284</u>	<u>₩ 17,777,338</u>

¹ Additions to the right-of-use assets during the year ended December 31, 2019 were ₩ 1,757,437 thousand.

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The amounts relating to leases recognized for the year ended December 31, 2019, on the statement of profit or loss, are as follows:

<i>(in thousands of Korean won)</i>		2019
Statement of profit or loss:		
Depreciation of right-of-use assets		
Buildings	₩	4,358,579
Machinery		3,964,354
Vehicle		51,681
		<u>342,544</u>
Interest expenses		521,337
Expense relating to variable lease payments not included in lease liabilities		2,091,747
Expense relating to leases of low-value assets		240,624
	₩	<u>7,212,287</u>

Classification of depreciation of right-of-use assets for the year ended December 31, 2019, is as follows:

<i>(in thousands of Korean won)</i>		2019
Cost of revenue	₩	3,632,110
Selling, general and administrative expenses		726,470
	₩	<u>4,358,580</u>

The total cash outflow from leases in relation to lease liabilities, low-value assets and variable lease payments during the year ended December 31, 2019 was ₩ 6,500,158 thousand.

21. Other Liabilities

Details of other liabilities as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019		2018	
	Current	Non-current	Current	Non-current
Advance receipts	₩ 141,361,599	₩ 6,011,768	₩ 85,239,342	₩ -
Withholdings	<u>3,345,171</u>	<u>-</u>	<u>2,456,272</u>	<u>-</u>
	<u>₩ 144,706,770</u>	<u>₩ 6,011,768</u>	<u>₩ 87,695,614</u>	<u>₩ -</u>

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22. Employee Benefit

Recognized liabilities for employee benefits as at December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)

		2019		2018	
Defined benefit liabilities	Present value of defined benefit obligations	₩	67,189,324	₩	49,342,672
	Fair value of plan assets		(52,760,029)		(36,076,367)
			<u>14,429,295</u>		<u>13,266,305</u>
Trade and other payables	Accrued expenses ¹		32,876,570		25,410,702
	Long-term accrued expenses ¹		3,337,749		3,988,729
			<u>36,214,319</u>		<u>29,399,431</u>
		₩	<u>50,643,614</u>	₩	<u>42,665,736</u>

¹ The amounts include the accrued compensation for annual leaves, performance bonuses, and others payable to employees.

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Changes in defined benefit liabilities for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	Defined benefit obligations		Plan assets		Net defined benefit liabilities	
	2019	2018	2019	2018	2019	2018
Beginning balance	₩49,342,672	₩33,888,400	₩ (36,076,367)	₩ (29,892,336)	₩13,266,305	₩ 3,996,064
Included in profit or loss:						
- Current service cost	14,332,455	11,487,263	-	-	14,332,455	11,487,263
- Interest expense (income)	<u>1,433,440</u>	<u>1,130,987</u>	<u>(1,055,461)</u>	<u>(1,019,102)</u>	<u>377,979</u>	<u>111,885</u>
	<u>15,765,895</u>	<u>12,618,250</u>	<u>(1,055,461)</u>	<u>(1,019,102)</u>	<u>14,710,434</u>	<u>11,599,148</u>
Included in other comprehensive income or loss:						
- Remeasurement loss (gain)						
Actuarial loss arising from						
Demographic assumptions	1,713,371	-	-	-	1,713,371	-
Financial assumptions	(1,617,639)	2,036,754	-	-	(1,617,639)	2,036,754
Experience adjustment	4,413,247	2,561,008	-	-	4,413,247	2,561,008
- Loss on plan assets excluding interest income	<u>-</u>	<u>-</u>	<u>488,142</u>	<u>587,075</u>	<u>488,142</u>	<u>587,075</u>
	<u>4,508,979</u>	<u>4,597,762</u>	<u>488,142</u>	<u>587,075</u>	<u>4,997,121</u>	<u>5,184,837</u>
Others:						
- Contributions to plan assets	-	-	(17,186,203)	(6,638,266)	(17,186,203)	(6,638,266)
- Benefit payments	(3,006,859)	(2,617,301)	1,648,497	1,711,248	(1,358,362)	(906,053)
- Transfer from affiliated companies	<u>578,637</u>	<u>855,561</u>	<u>(578,637)</u>	<u>(824,986)</u>	<u>-</u>	<u>30,575</u>
	<u>(2,428,222)</u>	<u>(1,761,740)</u>	<u>(16,116,343)</u>	<u>(5,752,004)</u>	<u>(18,544,565)</u>	<u>(7,513,744)</u>
Ending balance	<u>₩67,189,324</u>	<u>₩49,342,672</u>	<u>₩ (52,760,029)</u>	<u>₩ (36,076,367)</u>	<u>₩14,429,295</u>	<u>₩13,266,305</u>

Details of plan assets as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Cash and cash equivalents	₩ 52,760,029	₩ 36,076,367

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The expenses recognized as profit or loss related to defined benefit obligation for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019		2018	
Cost of sales	₩	12,649,970	₩	8,409,227
Selling, general and administrative expenses		2,060,464		1,661,657
Construction-in-progress		-		1,528,264
	₩	<u>14,710,434</u>	₩	<u>11,599,148</u>

The significant actuarial assumptions as at December 31, 2019 and 2018, are as follows:

<i>(in percentage, %)</i>	2019		2018	
Discount rate	3.38	%	3.10	%
Future salary increases	6.01	%	6.05	%

As at December 31, 2019 and 2018, the weighted average duration of the defined benefit obligation is 8.91 years and 8.1 years, respectively.

As at December 31, 2019, effects on present value of defined benefit obligations if each of significant actuarial assumptions changes within potential reasonable range are as follows (if changed by 1%):

<i>(in thousands of Korean won)</i>	Salary increases		Discount rate	
	1% Up	1% Down	1% Up	1% Down
Present value of defined benefit obligations	₩ 6,735,070	₩ (5,796,924)	₩ (5,683,266)	₩ 6,720,688

23. Share Capital and Reserves

Details of share capital as at December 31, 2019 and 2018, are summarized as follows:

<i>(in Korean won, in shares)</i>	2019		2018	
Number of shares authorized		500,000,000		500,000,000
Par value	₩	2,500	₩	2,500
Number of shares issued		66,165,000		66,165,000
Share capital	₩	165,412,500,000	₩	165,412,500,000

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Details of share premium as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Additional paid-in capital	₩ 2,496,583,409	₩ 2,496,583,409
Disproportionate capital increase	<u>(9,270,327)</u>	<u>(9,270,327)</u>
	<u>₩ 2,487,313,082</u>	<u>₩ 2,487,313,082</u>

Details of accumulated other comprehensive loss as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Remeasurements of net defined benefit liabilities	₩ (16,328,044)	₩ (12,540,227)
Share of other comprehensive income of associates and joint ventures	<u>5,433,278</u>	<u>5,564,266</u>
	<u>₩ (10,894,766)</u>	<u>₩ (6,975,961)</u>

The appropriation of retained earnings for the year ended December 31, 2019, is expected to be appropriated at the shareholders' meeting on March 20, 2020. The appropriation date for the year ended December 31, 2018, was March 22, 2019.

The appropriation of retained earnings for the years ended December 31, 2019 and 2018, is as follows:

<i>(in Korean won)</i>	2019	2018
Unappropriated retained earnings carried over from prior year	₩ 1,509,718,830,210	₩ 1,327,734,757,202
Adjustment on initial application of Korean IFRS 1115	-	(41,974,908,757)
Share of other comprehensive income of associates and joint ventures	-	(150,213,107)
Profit for the year	<u>202,904,099,088</u>	<u>224,109,194,872</u>
Retained earnings available for appropriation	<u>1,712,622,929,298</u>	<u>1,509,718,830,210</u>
Appropriation of retained earnings	-	-
Unappropriated retained earnings to be carried forward	<u>₩ 1,712,622,929,298</u>	<u>₩ 1,509,718,830,210</u>

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24. Revenue and Cost of Revenue

Details of revenue and cost of revenue for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Revenue:		
Product sales	₩ 648,374,697	₩ 528,227,493
Other sales	53,217,163	7,578,308
	<u>₩ 701,591,860</u>	<u>₩ 535,805,801</u>
Cost of revenue:		
Cost of product sales	₩ 371,796,917	₩ 346,381,759
Cost of other sales ¹	137,787,508	44,264,100
	<u>₩ 509,584,425</u>	<u>₩ 390,645,859</u>

¹ Cost of other sales includes loss on abnormal capacity and provision for valuation of inventories.

25. Selling, General and Administrative Expenses

Details of selling, general and administrative expenses for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Salaries	₩ 17,638,698	₩ 13,867,455
Bonuses	6,629,303	8,985,853
Retirement benefits	2,060,464	1,661,657
Employee welfare	6,189,505	5,655,411
Depreciation	5,712,650	11,791,680
Amortization of intangible assets	3,539,125	2,823,704
Repairs and maintenance	475,603	132,180
Supplies expenses	1,429,424	921,345
Utility expenses	227,280	221,958
Travel expenses	1,577,942	1,501,657
Service fees	28,300,374	20,541,340
Training expenses	6,666,629	5,795,721
Electronic processing expenses	2,109,119	1,890,192
Taxes and dues	1,659,938	1,564,795
Insurance premium	3,574,292	3,636,748
Advertising expenses	6,480,625	4,422,807
Others	5,994,076	4,066,694
	<u>₩ 100,265,047</u>	<u>₩ 89,481,197</u>

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26. Expenses by Nature

Details of expenses classified by nature for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019		2018	
Changes in work-in-process and finished goods	₩	(108,799,281)	₩	(35,822,261)
Changes in costs to fulfill a contract		(54,103,717)		(53,591,271)
Raw materials and supplies used		175,625,799		154,164,249
Employee benefits		232,214,318		169,729,301
Depreciation and amortization		128,526,523		91,450,555
Supplies expenses		56,664,072		36,881,951
Utility expenses		31,152,625		21,208,597
Service fees		52,063,213		35,680,563
Others		96,505,921		60,425,373
Total cost of sales, and selling, general and administrative expenses	₩	<u>609,849,473</u>	₩	<u>480,127,057</u>

27. Other Income and Expenses

Details of other income for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019		2018	
Rental revenues	₩	1,367,525	₩	497,992
Fees revenues		139,790		166,509
Gain on disposal of property, plant and equipment		520		12,821
Gain on disposal of investments in associates		-		389,213,672
Gain on lease cancellation		117		-
Miscellaneous gains		2,495,535		508,131
	₩	<u>4,003,487</u>	₩	<u>390,399,125</u>

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Details of other expenses for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Donations	₩ 1,000	₩ 13,000
Loss on disposal of property, plant and equipment	312,537	614,509
Loss on disposal of intangible assets	197,567	10,756
Loss on retirement of property, plant and equipment	296,625	-
Impairment loss on intangible assets	359,600	-
Miscellaneous losses	941,452	8,110,008
	<u>₩ 2,108,781</u>	<u>₩ 8,748,273</u>

28. Finance Income and Costs

Details of finance income for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Interest income	₩ 17,246,464	₩ 9,651,186
Gain on valuation of financial assets at fair value through profit or loss	45,759	80,517
Gain on disposal of financial assets at fair value through profit or loss	361,459	301,927
Gain on valuation of derivative assets	1,193,321	-
Gain on foreign currency translations	143,426	112,646
Gain on foreign currency transactions	8,550,315	9,174,513
	<u>₩ 27,540,744</u>	<u>₩ 19,320,789</u>

Details of finance costs for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Interest expenses	₩ 25,614,722	₩ 15,838,589
Loss on valuation of derivative liabilities	-	190,330
Loss on foreign currency translations	7,436,127	3,291,913
Loss on foreign currency transactions	5,631,794	5,016,385
	<u>₩ 38,682,643</u>	<u>₩ 24,337,217</u>

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29. Income Tax Expense (Benefit)

Income tax expense (benefit) for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Current income tax expense:		
Current income tax	₩ 15,377,052	₩ 40,772,131
Adjustments	(38,759)	-
Deferred tax expense:		
Temporary differences	(63,997,415)	29,226,438
Income taxes recognized directly in equity	1,190,332	8,868,073
Income tax expense (benefit)	<u>₩ (47,468,790)</u>	<u>₩ 78,866,642</u>

The relationship between tax expense (benefit) and accounting profit for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Profit before income tax	₩ 155,435,309	₩ 302,975,837
Income taxes at statutory tax rate	37,615,345	73,320,153
Adjustments:		
Non-deductible expense	260,810	2,161,758
Changes in unrecognized deferred tax assets	(82,596,726)	9,015,013
Tax credit carryforward	(2,201,242)	(4,940,694)
Others	(546,977)	(689,588)
Income tax expense (benefit)	<u>₩ (47,468,790)</u>	<u>₩ 78,866,642</u>
Average effective tax rate ¹	-	26.03%

¹ The effective tax rate for the year ended December 31, 2019, has not been calculated since the effective tax rate is negative.

Income taxes recognized directly in other comprehensive income (loss) for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019			2018		
	Before tax	Tax expense	Net of tax	Before tax	Tax expense	Net of tax
Defined benefit plan						
remeasurements	₩ (4,997,122)	₩ 1,209,304	₩(3,787,818)	₩(5,184,837)	₩1,254,731	₩(3,930,106)
Equity adjustments arising from						
investments in associates and						
joint ventures	(112,014)	(18,972)	(130,986)	963,592	-	963,592
	<u>₩ (5,109,136)</u>	<u>₩ 1,190,332</u>	<u>₩(3,918,804)</u>	<u>₩(4,221,245)</u>	<u>₩1,254,731</u>	<u>₩(2,966,514)</u>

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Income taxes recognized directly in equity for the years ended December 31, 2019 and 2018 are as follows:

(in thousands of Korean won)	2019			2018		
	Before tax	Tax effect	Net of tax	Before tax	Tax effect	Net of tax
Retained earnings ¹	₩	- ₩	- ₩	- ₩ (49,588,251)	₩7,613,342	₩(41,974,909)

¹ This occurred as a result of adjusting the cumulative effect of applying Korean IFRS 1115 *Revenue from contracts with customers*, to the equity as at January 1, 2018.

Changes in temporary differences and deferred tax assets (liabilities) as at December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)	2019			
	January 1, 2019	Profit or loss	Other comprehensive income	December 31, 2019
Accrued expenses	₩ 5,167,300	₩ 1,634,361	₩ -	₩ 6,801,661
Net defined benefit obligations	2,994,582	(739,166)	1,209,304	3,464,720
Loss (gain) on foreign currency translations	185,205	1,965,393	-	2,150,598
Provision for loss on valuation of finished goods	1,559,171	306,807	-	1,865,978
Provision for loss on valuation of work-in-progress	102,777	1,197,316	-	1,300,093
Depreciation	2,895,795	(59,754)	-	2,836,041
Provision for performance bonus	4,312,416	1,068,076	-	5,380,492
Gain from assets contributed	-	696,975	-	696,975
Investments in associates and joint ventures	(581,125,775)	64,991,298	(18,972)	(516,153,449)
Accrued income	(389,753)	100,407	-	(289,346)
Emission liability	361,544	(227,511)	-	134,033
Contract assets	(23,372,031)	(13,286,828)	-	(36,658,859)
Contract liabilities	24,467,360	10,653,921	-	35,121,281
Lease liabilities / Right-of-use assets	-	65,143	-	65,143
Impairment of intangible assets	2,603	87,023	-	89,626
Gain on valuation of derivative assets-SWAP	46,060	(288,784)	-	(242,724)
Tax credit carryforward	28,250,416	(5,357,594)	-	22,892,822
Deferred tax assets (liabilities)	₩ (534,542,330)	₩ 62,807,083	₩ 1,190,332	₩ (470,544,915)

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won)

	2018					
	January 1, 2018	Adjustments of beginning	Profit or loss	Other comprehensive income	December 31, 2018	
Accrued expenses	₩ 3,327,353	₩ -	₩ 1,839,947	₩ -	₩ 5,167,300	
Net defined benefit obligations	949,957	-	789,894	1,254,731	2,994,582	
Loss (gain) on foreign currency translations	(579,202)	-	764,407	-	185,205	
Provision for loss on valuation of finished goods	2,128,058	-	(568,887)	-	1,559,171	
Provision for loss on valuation of work-in-progress	1,638	-	101,139	-	102,777	
Depreciation	161,942	-	2,733,853	-	2,895,795	
Provision for performance bonus	3,862,496	-	449,920	-	4,312,416	
Derivative liabilities	467,922,107	-	(467,922,107)	-	-	
Investments in associates and joint ventures	(1,039,776,584)	3,913,698	454,737,111	-	(581,125,775)	
Accrued income	(15,446)	-	(374,307)	-	(389,753)	
Emission liability	121,503	-	240,041	-	361,544	
Contract assets	-	(10,200,929)	(13,171,102)	-	(23,372,031)	
Contract liabilities	-	13,900,573	10,566,787	-	24,467,360	
Impairment of intangible assets	-	-	2,603	-	2,603	
Gain on valuation of derivative assets-SWAP	-	-	46,060	-	46,060	
Tax loss carryforward	33,270,565	-	(33,270,565)	-	-	
Tax credit carryforward	23,309,722	-	4,940,694	-	28,250,416	
Deferred tax assets (liabilities)	₩ (505,315,891)	₩ 7,613,342	₩ (38,094,512)	₩ 1,254,731	₩ (534,542,330)	

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The analysis of deferred tax assets and liabilities as at December 31, 2019 and 2018, is as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Deferred tax assets		
Deferred tax asset to be recovered after more than 12 months	₩ 59,872,986	₩ 53,573,078
Deferred tax asset to be recovered within 12 months	22,926,477	16,772,151
	<u>82,799,463</u>	<u>70,345,229</u>
Deferred tax liabilities		
Deferred tax liability to be recovered after more than 12 months	(547,775,720)	(599,779,005)
Deferred tax liability to be recovered within 12 months	(5,568,658)	(5,108,554)
	<u>(553,344,378)</u>	<u>(604,887,559)</u>
Deferred tax assets (liabilities), net	<u>₩ (470,544,915)</u>	<u>₩ (534,542,330)</u>

As at December 31, 2019 and 2018, the Company did not recognize ₩ 33,339,136 thousand and ₩ 115,935,862 thousand of deferred tax assets, respectively, with respect to investments in associates and joint ventures, tax credit carryforward, and so forth as the Company determines that it is not probable that they will be realized in foreseeable future.

Realization of the future tax benefits related to the deferred tax assets is dependent on various factors, including the Company's ability to generate taxable income within the period during which the temporary differences reverse, the outlook of the Korean economic environment, and the overall future industry outlook. Management periodically considers these factors in reaching its conclusion and recognizes deferred income tax assets based on the possibility of future realization.

During the year ended December 31, 2019, the Company recognized deferred tax assets of ₩ 64,972,325 thousand, which were not recognized in relation to the temporary differences from investments in associates during the prior year.

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30. Earnings per Share

Basic earnings per share for the years ended December 31, 2019 and 2018, is calculated as follows:

<i>(in Korean won)</i>	2019	2018
Profit for the year	₩ 202,904,099,088	₩ 224,109,194,872
Weighted-average number of ordinary shares outstanding ¹	<u>66,165,000</u>	<u>66,165,000</u>
Basic earnings per share	<u>₩ 3,067</u>	<u>₩ 3,387</u>

¹ There has been no change in weighted-average number of ordinary shares outstanding during for the years ended December 31, 2019 and 2018.

The Company has no dilutive potential shares; therefore, diluted earnings per share is equal to the basic earnings per share.

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31. Commitments and Contingencies

As at December 31, 2019, details of the Company's borrowings commitment with financial institutions are summarized as follows:

(in thousands of Korean won and in thousands of USD)

	Type	Currency	Limit	Currency	Outstanding balance
Woori Bank	Bank overdraft	KRW	10,000,000	KRW	-
	B2B PLUS	KRW	5,760,000	KRW	-
	General loans for operation	KRW	30,000,000	KRW	-
	Loans for equipment	KRW	70,000,000	KRW	70,000,000
KEB Hana Bank	Bank overdraft	KRW	10,000,000	KRW	-
	Loans for equipment	KRW	50,000,000	KRW	50,000,000
	Loans for equipment	KRW	49,700,000	KRW	500,000
	Comprehensive limit	KRW	40,000,000	KRW	-
	General loan in foreign currency	USD	30,000	USD	30,000
Shinhan Bank	General loan	KRW	19,214,000	KRW	-
	Bank overdraft	KRW	10,000,000	KRW	-
	Loans for equipment	KRW	70,000,000	KRW	70,000,000
Kookmin Bank	Loans for equipment	KRW	18,750,000	KRW	18,750,000
	Loans for equipment	KRW	60,000,000	KRW	60,000,000
Korea Development Bank	Loans for equipment	KRW	12,500,000	KRW	12,500,000
	Loans for equipment	KRW	20,000,000	KRW	20,000,000
Mizuho Bank	Loans for equipment	KRW	70,000,000	KRW	70,000,000
Sumitomo Mitsui Bank	General loan	KRW	30,000,000	KRW	30,000,000
	Overseas investment loans	USD	10,000	USD	10,000

As at December 31, 2019, details of guarantees provided by financial institutions are summarized as follows:

(in thousands of Korean won and in thousands of USD)

	Remarks	Currency	Limit	Currency	Outstanding balance
Woori Bank	Payment guarantees	KRW	9,500,000	KRW	2,071,579
Shinhan Bank	Letter of credit	USD	10,000	USD	-
	Foreign currency payment guarantee	USD	10,000	USD	-
Kookmin Bank	Foreign currency payment guarantee	KRW	30,000,000	USD	-

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If the Company does not comply with the commitments below, financial institutions may require an early redemption of long-term borrowings. Details of debt covenants related to long-term borrowings with financial institutions as at December 31, 2019, are as follows:

Financial institution	Remarks	Details of covenants
Kookmin Bank	Retention of management rights	The Company shall continue to be an affiliate of the Samsung Group, and a current largest shareholder should retain its management rights.
Sumitomo Mitsui Bank	Shareholder composition ratio	The ownership of the Samsung Group shall be over 50%, and Samsung C&T Corporation shall maintain its ownership in the Company in excess of 30%.
Mizuho Bank	Shareholder composition ratio	Samsung Group (Samsung C&T Corporation, Samsung Electronics) shall maintain ownership of the Company in excess of 60% (voting shares).

In connection with a lease agreement with Incheon Metropolitan city ("Incheon") related to the use of land, the Company is exempt from the full amount of rent because the Company meets the requirements for exemption of rent in accordance with the *Common Property Management Ordinance of Incheon*. The Company recognized an intangible asset at the fair value of the rent that is exempt from Incheon and government grant as a reduction of assets at the same value. The period of land use without payment and exemption requirement of the full amount of rent are as follows:

(i) Period

Lease terms are for a total of 50 years. The initial lease period is for 20 years from the land lease contract date, April 28, 2011, and the Company can renew the contract for another 30 years in successive 10 year increments.

(ii) Exemption requirement

The Company whose foreign investment amounts to more than USD 20 million for 5 years since the land lease contract date is subject to the exemption from Article 32 of *Common Property Management Ordinance of Incheon*. Afterwards, the Company, including foreign invested enterprises that meet the average daily employee count of above 300, is subject to exemption.

On April 29, 2019, a group of minority shareholders has launched a collective lawsuit against the Company, claiming ₩ 8,449 million of damages of investment losses due to accounting treatment for investments on Samsung Bioepis Co. Ltd. (Note 35) and the litigation is in progress.

Contractual obligation to purchase property, plant and equipment and intangible assets as at December 31, 2019 are ₩ 17,396 million.

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32. Related Parties

As at December 31, 2019, Samsung C&T Corporation is a controlling company, and the percentage of ownership it has over the Company is 43.44%.

Details of related parties as at December 31, 2019, are as follows:

Relationship	Companies
Parent company	Samsung C&T Corporation
Entity with significant influence over the Company	Samsung Electronics Co., Ltd.
Associates and Joint ventures	Samsung Bioepis Co., Ltd., Archigen Biotech Limited
Others	Samsung Welstory Inc., Samsung C&T Deutschland GmbH, Samsung Research America, Inc, Samsung Electronics Sales Co., Ltd.
The Enterprise Group ¹	Samsung Card Co., Ltd., Samsung Engineering Co., Ltd. S-1 Corporation, Samsung SDS Co., Ltd., Samsung Economic Research Institute, Multicampus Corporation, Samsung Fire & Marine Insurance Co., Ltd., Samsung Life Insurance Co., Ltd., Hotel Shilla Co., Ltd., Cheil Worldwide Inc., Samsung Securities Co.,Ltd, SBTM Corporation and others

¹ Although the entities are not the related party of the Company in accordance with Korean IFRS 1024, the entities belong to the Business Group designated as a conglomerate to which the Company also belongs in accordance with the Monopoly Regulation and Fair Trade Act.

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Outstanding balances with related parties as at December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)	2019					
	Receivables			Payables		
	Trade receivables	Other receivables	Accrued income	Other payables and others	Advance receipts	Contract liabilities
Entity with significant influence over the Company						
Samsung Electronics Co., Ltd.	₩ -	₩ -	₩ -	₩ 3,148,703	₩ -	₩ -
Associates and joint ventures						
Samsung Bioepis Co., Ltd.	98,332,044	154,500	575,929	490,119	40,812,620	36,238,155
Archigen Biotech Limited	-	-	177,480	-	-	-
Others						
Samsung Welstory Inc.	-	3,484	-	765,000	-	-
Samsung C&T Deutschland GmbH	-	-	-	28,266	-	-
Samsung Research America, Inc.	-	-	-	90,675	-	-
The Enterprise Group						
Samsung Card Co., Ltd.	-	29,418	-	624,811	-	-
S-1 Corporation	-	-	-	2,684,430	-	-
Samsung SDS Co., Ltd.	-	-	-	1,324,628	-	-
Samsung Economic Research Institute	-	-	-	78,816	-	-
Samsung Fire & Marine Insurance Co., Ltd.	-	54	-	3,687	-	-
Others ¹	-	-	-	253,171	-	-
	<u>₩98,332,044</u>	<u>₩ 187,456</u>	<u>₩ 753,409</u>	<u>₩ 9,492,306</u>	<u>₩40,812,620</u>	<u>₩36,238,155</u>

¹ Transactions with Multicampus Corporation, Hotel Shilla Co., Ltd., Cheil Worldwide Inc. and SBTM Corporation were included.

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	2018									
	Receivables					Payables				
	Trade receivables	Other receivables	Accrued income	Other payables and others	Advance receipts	Contract liabilities				
Entity with significant influence over the Company										
Samsung Electronics Co., Ltd.	₩ -	₩ -	₩ -	₩ 1,777,735	₩ -	₩ -				
Associates and joint ventures										
Samsung Bioepis Co., Ltd.	65,408,494	106,572	522,506	-	2,077,939	18,289,073				
Archigen Biotech Limited	-	-	277,371	-	-	-				
Others										
Samsung Welstory Inc.	-	3,014	-	727,587	-	-				
Samsung Research America, Inc.	-	-	-	94,194	-	-				
Samsung C&T Deutschland GmbH	-	-	-	80,785	-	-				
The Enterprise Group										
Samsung Card Co., Ltd.	-	11,979	-	253,644	-	-				
Samsung Engineering Co., Ltd.	-	-	-	75,343,455	-	-				
S-1 Corporation	-	-	-	1,370,857	-	-				
Samsung SDS Co., Ltd.	-	-	-	2,621,992	-	-				
Samsung Economic Research Institute	-	-	-	322,761	-	-				
Samsung Fire & Marine Insurance Co., Ltd.	-	-	-	960	-	-				
Others ¹	-	-	-	115,258	-	-				
	<u>₩65,408,494</u>	<u>₩ 121,565</u>	<u>₩ 799,877</u>	<u>₩82,709,228</u>	<u>₩ 2,077,939</u>	<u>₩18,289,073</u>				

¹ Transactions with Multicampus Corporation and Hotel Shilla Co., Ltd. were included.

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Significant transactions which occurred in the normal course of business with related parties for the years ended December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)

	2019				
	Revenue		Expenses		Asset acquisition
	Revenue	Others	Service fees	Others	
Parent company					
Samsung C&T Corporation	₩	-	₩	-	₩ 301,716
Entity with significant influence over the Company					
Samsung Electronics Co., Ltd.		-		-	2,920,609
Associates and joint ventures					
Samsung Bioepis Co., Ltd. ¹	183,125,111	1,426,113	-	499,686	20,163,000
Archigen Biotech Limited	-	614,999	-	-	-
Others					
Samsung Welstory Inc.	-	34,365	-	8,408,532	-
Samsung Electronics Sales Co., Ltd.	-	3,600	-	18,020	-
Samsung C&T Deutschland GmbH	-	-	-	1,015,612	-
Samsung Research America, Inc.	-	-	1,318,229	-	-
The Enterprise Group					
Samsung Card Co., Ltd.	-	21,312	-	317,249	-
S-1 Corporation	-	-	12,960,138	2,160,667	1,631,122
Samsung SDS Co., Ltd.	-	2,666	516,092	12,680,064	2,156,950
Samsung Economic Research Institute	-	-	180,000	1,586,878	-
Samsung Fire & Marine Insurance Co., Ltd. ²	-	2,545,322	2,858	4,173,889	-
Samsung Life Insurance Co., Ltd. ²	-	143,959	-	36,641	-
Samsung Securities Co., Ltd. ²	-	136,152	27,849	-	-
Cheil Worldwide Inc.	-	-	-	158,337	-
SBTM Corporation	-	-	5,815	18,825	-
Others ³	-	-	17,674	1,014,566	-
	₩183,125,111	₩ 4,928,488	₩ 15,330,371	₩ 35,159,575	₩ 23,951,072

¹ Acquired assets are leased to Samsung Bioepis Co., Ltd (Note 14).

² Other revenue includes actual return on plan assets recognized as deduction of retirement benefits. As at December 31, 2019, the Company deposited ₩ 27,888 million (2018: ₩ 22,862 million) in Samsung Fire & Marine Insurance Co., Ltd., ₩ 12,427 million (2018: ₩ 6,606 million) in Samsung Life Insurance Co., Ltd., and ₩ 12,445 million (2018: ₩ 6,609 million) in Samsung Securities Co., Ltd., as plan assets. The Company contributed additional amount of ₩ 5,786 million, ₩ 5,700 million, and ₩ 5,700 million, respectively, for the year ended December 31, 2019.

³ Transactions with Multicampus Corporation and Hotel Shilla Co., Ltd. were included.

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	2018				
	Revenue		Expenses		Asset acquisition
	Revenue	Others	Service fees	Others	
Parent company					
Samsung C&T Corporation	₩	-	₩	363,021	₩ 3,280,317
Entity with significant influence over the Company					
Samsung Electronics Co., Ltd.		-	-	1,621,550	-
Associates and joint ventures					
Samsung Bioepis Co., Ltd.	225,444,041	591,158	-	-	-
Archigen Biotech Limited	-	17,695	-	9,384	-
Others					
Samsung Welstory Inc.	-	29,798	-	7,426,091	-
Samsung C&T Deutschland GmbH	-	-	-	800,507	-
Samsung Electronics Sales Co., Ltd.	-	3,600	-	33,368	-
Samsung Research America, Inc.	-	-	1,758,213	-	-
The Enterprise Group					
Samsung Card Co., Ltd.	-	7,902	-	313,846	-
Samsung Engineering Co., Ltd.	-	-	-	-	121,129,703
S-1 Corporation	-	-	11,114,727	929,556	810,600
Samsung SDS Co., Ltd.	-	-	6,489	12,525,616	3,652,816
Samsung Economic Research Institute	-	-	-	1,320,090	-
Samsung Fire & Marine Insurance Co., Ltd. ¹	-	412,335	-	3,909,654	-
Samsung Life Insurance Co., Ltd. ¹	-	81,300	-	27,877	-
Others ²	-	-	38,352	763,824	-
	<u>₩225,444,041</u>	<u>₩ 1,143,788</u>	<u>₩ 13,280,802</u>	<u>₩ 29,681,363</u>	<u>₩128,873,436</u>

¹ Other revenue includes actual return on plan assets recognized as deduction of retirement benefits.

² Transactions with Multicampus Corporation and Hotel Shilla Co., Ltd. were included.

The compensation paid or payable to key management for employee services for the years ended December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)

	2019	2018
Short-term employee benefits	₩ 2,128,248	₩ 3,419,030
Long-term employee benefits ¹	817,450	1,889,461
Retirement benefits	364,914	243,788
	<u>₩ 3,310,612</u>	<u>₩ 5,552,279</u>

¹ For the year ended December 31, 2018, long-term employee benefits include ₩ 612,892 thousand in unregistered director benefits.

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Key management refers to the registered directors. Compensation of the Company's key management personnel includes salaries, non-cash benefits, and contributions to a post-employment defined benefit plan. In addition, the Company has granted long-term incentives to the Company's key management personnel based on its three-year management performance. The estimated amount of liabilities arising from such incentives are recognized as accrued expenses and long-term accrued expenses.

Financing transactions with related parties for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	Companies	2019	2018	Description
Associates and joint ventures	Archigen Biotech Limited	₩ 18,101,000	₩ 16,405,000	Equity contribution in cash

33. Statements of Cash Flows

Significant transactions not involving cash flows of investing and financing activities for the years ended December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Increase in other payables due to acquisition of property, plant and equipment and intangible assets	₩ 15,158,486	₩ 80,691,934
Transfer construction-in-progress to property, plant and equipment and intangible assets	129,520,360	533,223,340

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Changes in liabilities arising from financial activities for the years ended December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)

	2019						
	Current portion of debentures	Debentures	Short-term borrowings	Current portion of long-term borrowings	Long-term borrowings	Lease liabilities	Total
At January 1, 2019	₩ 50,000,000	₩ 189,638,752	₩ 90,000,000	₩ 246,583,355	₩ 296,474,000	₩ 17,777,338	₩ 890,473,445
Changes arising from financial activities:							
Proceeds from borrowing liabilities	-	-	150,000,000	-	-	-	150,000,000
Repayment of borrowing liabilities	-	-	(90,000,000)	(246,583,355)	-	-	(336,583,355)
Repayment of debentures issued	(50,000,000)	-	-	-	-	-	(50,000,000)
Payment of lease	-	-	-	-	-	(3,222,220)	(3,222,220)
Others:							
Reclassification to current portion	-	-	-	251,650,000	(251,650,000)	-	-
Effects of exchange rate changes on cash and cash equivalents	-	-	-	-	1,588,000	-	1,588,000
Amortization of discount on debentures	-	116,373	-	-	-	-	116,373
Changes in lease	-	-	-	-	-	1,334,282	1,334,282
Gain on lease cancellation	-	-	-	-	-	(117)	(117)
At December 31, 2019	₩ -	₩ 189,755,125	₩ 150,000,000	₩ 251,650,000	₩ 46,412,000	₩ 15,889,283	₩ 653,706,408

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2018

	Current portion of debentures		Debentures		Short-term borrowings		Current portion of long-term borrowings		Long-term borrowings		Total
At January 1, 2018	₩	-	₩	99,926,578	₩	40,000,000	₩	48,750,000	₩	357,235,570	₩ 545,912,148
Changes arising from financial activities:											
Proceeds from borrowing liabilities		-		-		50,000,000		-		153,112,000	203,112,000
Proceeds from debentures issued		-		189,579,060		-		-		-	189,579,060
Repayment of borrowing liabilities		-		-		-		(18,750,000)		-	(18,750,000)
Repayment of debentures issued		-		(50,000,000)		-		-		-	(50,000,000)
		-		139,579,060		50,000,000		(18,750,000)		153,112,000	323,941,060
Reclassification to current portion		50,000,000		(50,000,000)		-		215,485,570		(215,485,570)	-
Effects of exchange rate changes on cash and cash equivalents		-		-		-		1,097,785		1,612,000	2,709,785
Amortization of discount on debentures		-		133,114		-		-		-	133,114
At December 31, 2018	₩	50,000,000	₩	189,638,752	₩	90,000,000	₩	246,583,355	₩	296,474,000	₩ 872,696,107

34. Financial Risk Management

The Company is exposed to credit risk, liquidity risk, and market risk in connection with its financial instruments. This note presents information about the Company's exposure to each of the above risks, the Company's objectives, policies and processes for measuring and managing risks, and the Company's management of capital. Further quantitative disclosures are included throughout these financial statements.

(a) Financial risk management

The Company's risk management policies are established to identify and analyze the risks faced by the Company, to set appropriate risk limits and controls, and to monitor risks and adherence to limits. Risk management policies and systems are reviewed regularly to reflect changes in market conditions and the Company's activities. The Company, through its training and management standards and procedures, aims to maintain a disciplined and constructive control environment in which all employees understand their roles and obligations. The Board of Directors oversees how management monitors compliance with the Company's risk management policies and procedures and reviews the adequacy of the risk management framework in relation to the risks faced by the Company.

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(b) Credit risk

Credit risk is the risk of financial loss to the Company if a customer or counterparty to a financial instrument fails to meet its contractual obligations and arises principally from the Company's receivables from customers. The carrying amount of financial assets represents the maximum credit exposure. The maximum exposures to credit risk as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>	2019	2018
Cash and cash equivalents	₩ 125,100,653	₩ 17,621,905
Short-term financial instruments	320,000,000	550,000,000
Loans	400,000	600,000
Financial assets at fair value through profit or loss	210,040,293	540,080,517
Long-term financial instruments	7,000	7,000
Trade and other receivables	187,118,116	121,733,350
Derivative assets	1,002,990	-
Deposits provided	42,877,973	44,509,965
	<u>₩ 886,547,025</u>	<u>₩ 1,274,552,737</u>

(c) Liquidity risk

Liquidity risk is the risk that the Company will encounter difficulty in meeting the obligations associated with its financial liabilities that are settled by delivering cash or other financial assets. The Company's approach to managing liquidity is to ensure, as far as possible, that it will always have sufficient liquidity to meet its liabilities when they are due, under both normal and stressed conditions, without incurring unacceptable losses or risking damage to the Company's reputation.

The ultimate responsibility for liquidity risk management remains with the Board of Directors, who establishes fundamental policies to adequately manage short-term, medium-and long-term financing and liquidity management regulations. To maintain adequate liquidity, the Company manages liquidity risk by periodically forecasting and adjusting its capital balance.

The following are the contractual maturities of financial liabilities as at December 31, 2019 and 2018, including estimated interest payments and excluding the impact of netting agreements. The Company does not expect that the cash flows included in the maturity analysis could occur significantly earlier or at different amounts.

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<i>(in thousands of Korean won)</i>	2019					
	Book amount	Contractual cash flows	Less than 1 year	1-2 year	2-3 year	More than 3 years
Trade and other payables	₩ 109,233,102	₩ 109,233,102	₩ 101,128,061	₩ 8,105,041	₩ -	₩ -
Lease liabilities	15,889,284	17,330,666	3,756,506	3,539,260	3,292,400	6,742,500
Debentures	189,755,125	200,840,489	6,073,394	152,901,425	1,412,800	40,452,870
Short-term borrowings	150,000,000	152,654,775	152,654,775	-	-	-
Long-term borrowings	298,062,000	307,852,511	258,340,729	13,006,681	1,252,814	35,252,287
	<u>₩ 762,939,511</u>	<u>₩ 787,911,543</u>	<u>₩ 521,953,465</u>	<u>₩ 177,552,407</u>	<u>₩ 5,958,014</u>	<u>₩ 82,447,657</u>

<i>(in thousands of Korean won)</i>	2018					
	Book amount	Contractual cash flows	Less than 1 year	1-2 year	2-3 year	More than 3 years
Trade and other payables	₩ 174,327,024	₩ 174,327,024	₩170,338,294	₩ 3,988,730	₩ -	₩ -
Debentures	239,638,752	258,311,904	57,471,415	6,073,394	152,901,425	41,865,670
Short-term borrowings	90,000,000	92,050,285	92,050,285	-	-	-
Long-term borrowings	543,247,685	567,874,119	262,062,238	259,015,013	12,654,638	34,142,230
	<u>₩1,047,213,461</u>	<u>₩1,092,563,332</u>	<u>₩581,922,232</u>	<u>₩269,077,137</u>	<u>₩165,556,063</u>	<u>₩ 76,007,900</u>

(d) Currency risk

The Company is exposed to currency risk on sales, purchases, and borrowings. The currencies in which these transactions primarily are denominated are USD and so forth. The Company establishes a plan to manage currency risk through currency forwards and so forth, if considered necessary.

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(i) Exposure to currency risk

The Company's exposure to foreign currency risk as at December 31, 2019 and 2018, are as follows:

(in thousands of Korean won)	2019				
	USD	EUR	CHF	JPY	GBP
Assets:					
Cash equivalents	₩ 115,780,000	₩ -	₩ -	₩ -	₩ -
Trade and other receivables	180,179,596	12,171	-	-	132,990
Contract assets	1,630,646	-	-	-	-
	<u>297,590,242</u>	<u>12,171</u>	<u>-</u>	<u>-</u>	<u>132,990</u>
Liabilities:					
Borrowings	11,578,000	-	-	-	-
Trade and other payables	2,347,447	229,693	22,530	2,324	105,546
	<u>13,925,447</u>	<u>229,693</u>	<u>22,530</u>	<u>2,324</u>	<u>105,546</u>
	<u>₩ 283,664,795</u>	<u>₩ (217,522)</u>	<u>₩ (22,530)</u>	<u>₩ (2,324)</u>	<u>₩ 27,444</u>

(in thousands of Korean won)	2018				
	USD	EUR	CHF	JPY	GBP
Assets:					
Cash equivalents	₩ -	₩ -	₩ -	₩ -	₩ -
Trade and other receivables	110,741,934	55,228	-	-	8,467
Contract assets	711,399	-	-	-	-
	<u>111,453,333</u>	<u>55,228</u>	<u>-</u>	<u>-</u>	<u>8,467</u>
Liabilities:					
Borrowings	37,464,355	-	-	-	-
Trade and other payables	1,377,411	3,671,545	108,852	20,264	62,078
	<u>38,841,766</u>	<u>3,671,545</u>	<u>108,852</u>	<u>20,264</u>	<u>62,078</u>
	<u>₩ 72,611,567</u>	<u>₩ (3,616,317)</u>	<u>₩ (108,852)</u>	<u>₩ (20,264)</u>	<u>₩ (53,611)</u>

¹ Foreign currency borrowings with a foreign currency swap are excluded as they are not exposed to currency risk.

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(ii) The significant exchange rates applied for the years ended December 31, 2019 and 2018, are as follows:

	2019				2018			
	Average rate		December 31, 2019		Average rate		December 31, 2018	
(in Korean won)								
USD	₩	1,165.65	₩	1,157.80	₩	1,100.30	₩	1,118.10
EUR		1,304.81		1,297.43		1,298.63		1,279.16
CHF		1,173.43		1,195.52		1,124.29		1,136.22
JPY(100¥)		1,069.76		1,063.47		996.27		1,013.18
GBP		1,487.83		1,518.69		1,468.22		1,420.32

(iii) Sensitivity analysis

The Company internally assesses the currency risk against changes in exchange rate fluctuations regularly. The table below summarizes the impact of weakened/strengthened Korean won on the Company's income before tax for the year. The analysis is based on the assumption that Korean won has weakened/strengthened by 10% with all other variables held constant.

	2019				2018			
	10% increase		10% decrease		10% increase		10% decrease	
(in thousands of Korean won)								
USD	₩	28,366,479	₩	(28,366,479)	₩	7,261,157	₩	(7,261,157)
EUR		(21,752)		21,752		(361,632)		361,632
CHF		(2,253)		2,253		(10,885)		10,885
JPY		(232)		232		(2,026)		2,026
GBP		2,744		(2,744)		(5,361)		5,361
	₩	28,344,986	₩	(28,344,986)	₩	6,881,253	₩	(6,881,253)

(e) Interest rate risk

The Company is exposed to interest rate risk because they raised funds with interests paid at variable rate. The effect of changes in interest rates by 0.5% applied to the profit and loss (before tax) for the years ended December 31, 2019 and 2018, are as follows:

	2019				2018			
	0.5% increase		0.5% decrease		0.5% increase		0.5% decrease	
(in thousands of Korean won)								
Interest expense	₩	(1,201,400)	₩	1,201,400	₩	(1,973,725)	₩	1,973,725

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(f) Capital management

The Company's capital management is for the purpose of maximizing the shareholders' interest by maintaining an optimal capital structure. To achieve optimal capital structure, the Company may improve financial structure monitoring financial ratios, such as debt-to-equity ratio.

The debt-to-equity ratio as at December 31, 2019 and 2018, are as follows:

<i>(in thousands of Korean won)</i>		2019		2018
Total liabilities (A)	₩	1,557,172,767	₩	1,824,940,197
Total equity (B)		4,354,453,745		4,155,468,451
Debt-to-equity ratio (A/B)		35.76%		43.92%

35. Ruling of the Securities and Futures Commission in relation to the accounting for investment in Samsung Bioepis Co., Ltd., and the related prosecution investigation

On July 12, 2018, the Securities & Futures Commission ("SFC") imposed the first administrative measures in accordance with the Act on External Audit of Stock Companies against the Company for alleged failure to disclose sufficient information in the Company's Financial Statements from 2012 to 2015 regarding the Company's joint venture agreement with Biogen Therapeutics Inc. The measures included i) compulsory designation of external auditors for three years; ii) recommendation of dismissal of the Company's executives in charge; and iii) prosecution against the Company and its representative director (CEO).

On November 14, 2018, the SFC imposed the second administrative measures in accordance with the Act on External Audit of Stock Companies against the Company for violation of accounting standards by inappropriately applying consolidated method regarding investments in Samsung Bioepis from 2012 to 2018. The measures included i) compulsory designation of external auditors for three years; ii) recommendation of dismissal of the Company's representative director (CEO) and executives in charge; iii) referral of the case against the Company and its representative director (CEO) to the Prosecutors Office; iv) retrospective restatement of financial statements; and v) imposition of administrative fine of ₩ 8 billion.

In order to demonstrate the legitimacy of the Company's accounting, the Company submitted requests for cancellation of the measures ordered by SFC on October 8, 2018 and November 27, 2018, and submitted requests for suspension of measures on November 27, 2018 and December 18, 2018, to the Seoul Administrative Court (the "Court"). The trial proceedings are in progress (as discussed more in the following paragraphs).

Upon the request to the Court on November 27, 2018 relating to the second measures (made on November 14, 2018), the Company received the decision from the Court to suspend the measures imposed by SFC, which include: i) an order of a corrective measure (retrospective

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restatement of financial statements); ii) compulsory designation of external auditors for three years; iii) recommendation of dismissal of the Company's representative director (CEO) and the executives in charge until 30 days after the Court makes its final ruling. The SFC appealed the Court's decision on January 30, 2019, but the Seoul High Court affirmed the decision of the Court and rejected the appeal made by the SFC on May 13, 2019. The SFC appealed to the Supreme Court on May 23, 2019, but the re-appeal was rejected by the Supreme Court on September 6, 2019.

In addition, regarding the request for suspension of the first measures imposed by the SFC (made in July 12, 2018) which were submitted on December 18, 2018, the Company received the decision from the Court to suspend measures ordered by the SFC on February 19, 2019, which include: i) external auditor designation by the regulator for three years; ii) recommendation of dismissal of the Company's representative director (CEO) and the executives in charge until 30 days after the Court makes its final ruling. The SFC appealed to the Seoul High Court for the decision on February 21, 2019, but the Seoul High Court affirmed the decision of the Court and rejected the appeal by the SFC on March 24, 2019. The SFC appealed to the Supreme Court on June 10, 2019, but the re-appeal was rejected by the Supreme Court on October 11, 2019.

Administrative lawsuits are in progress relating to the SFC's measures, and it is not possible to predict the outcome of the administrative lawsuits. However, if the Company loses the administrative lawsuits, the financial statements for 2012 and the subsequent reporting periods could potentially be restated. With respect to this matter, the investigation by the Seoul Central Prosecutors Office against the Company's CEO and CFO for alleged violation of the Act on External Audit of Stock Companies, the Financial Investment Services and Capital Market Act, the Act on the Aggregated Punishment, Etc. of Specific Economic Crimes, and the allegations of document destruction under Criminal Act is also in progress. It is not possible to predict the outcome of the investigation.

Meanwhile, on December 9, 2019, employees of the Company and Company's affiliate who were arrested on suspicion of evidence destruction, were found guilty of violating the Act on External Audit of Stock companies and Criminal Act in the first ruling of the Seoul Central District Court. All the employees of the Company's affiliates appealed the ruling above, and the employee of the Company who abandoned his or her appeal received suspended sentences for the destruction of evidence.

If the Company's financial statements are restated in accordance with the SFC's measures, the investment in associates, deferred tax liabilities (net), and the capital/retained earnings accounts as at December 31, 2019 would be decreased by ₩ 2,369,724 million, ₩ 470,545 million, ₩ 1,796,250 million respectively while the equity method profit for the year ended December 31, 2019 would be increased by ₩ 17,982 million, mainly due to the elimination of fair value valuation for Samsung Bioepis in 2015. As a result of this potential restatement, the total comprehensive income for the year ended December 31, 2019 would be increased by ₩ 10,290 million.

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(a) Impact on Financial Position

If the Company loses the administrative litigation, it shall restate the statements of financial position (consolidated statements of financial position from 2012 to 2014) as indicated by the Securities and Futures Commission and its impact shall be as follows.

(i) As at December 31, 2019 and 2018

(in thousands of Korean won)

	2019			2018		
	Before restatement	Adjustment	After restatement	Before restatement	Adjustment	After restatement
Assets						
Current assets	₩1,356,261,664	₩ -	₩1,356,261,664	₩1,577,683,716	₩ -	₩1,577,683,716
Non-current assets	4,555,364,848	(2,266,795,380)	2,288,569,468	4,402,724,932	(2,341,121,903)	2,061,603,029
Investments in associates and joint ventures	2,652,731,622	(2,369,723,569)	283,008,053	2,561,802,520	(2,387,705,348)	174,097,172
Deferred tax assets	-	102,928,189	102,928,189	-	46,583,445	46,583,445
Total assets	<u>₩5,911,626,512</u>	<u>₩ (2,266,795,380)</u>	<u>₩3,644,831,132</u>	<u>₩5,980,408,648</u>	<u>₩ (2,341,121,903)</u>	<u>₩3,639,286,745</u>
Liabilities						
Current liabilities	₩ 690,505,243	₩ -	₩ 690,505,243	₩ 710,529,787	₩ (38,759)	₩ 710,491,028
Current tax liabilities	12,857,398	-	12,857,398	39,623,718	(38,759)	39,584,959
Non-current liabilities	866,667,524	(470,544,915)	396,122,609	1,114,410,410	(534,542,329)	579,868,081
Deferred tax liabilities	470,544,915	(470,544,915)	-	534,542,330	(534,542,330)	-
Total liabilities	<u>1,557,172,767</u>	<u>(470,544,915)</u>	<u>1,086,627,852</u>	<u>1,824,940,197</u>	<u>(534,581,088)</u>	<u>1,290,359,109</u>
Equity						
Share premium	2,487,313,082	9,313,769	2,496,626,851	2,487,313,082	9,313,769	2,496,626,851
Accumulated other comprehensive loss	(10,894,766)	(1,738,979)	(12,633,745)	(6,975,961)	(1,738,979)	(8,714,940)
Retained earnings	1,712,622,929	(1,803,825,254)	(91,202,325)	1,509,718,830	(1,814,115,605)	(304,396,775)
Total equity	<u>4,354,453,745</u>	<u>(1,796,250,465)</u>	<u>2,558,203,280</u>	<u>4,155,468,451</u>	<u>(1,806,540,815)</u>	<u>2,348,927,636</u>
Total liabilities and equity	<u>₩5,911,626,512</u>	<u>₩ (2,266,795,380)</u>	<u>₩3,644,831,132</u>	<u>₩5,980,408,648</u>	<u>₩ (2,341,121,903)</u>	<u>₩3,639,286,745</u>

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(ii) As at December 31, 2017 and 2016

(in thousands of Korean won)

	2017			2016		
	Before restatement	Adjustment	After restatement	Before restatement	Adjustment	After restatement
Assets						
Current assets	₩ 620,995,135	₩ -	₩ 620,995,135	₩1,461,425,176	₩ -	₩1,461,425,176
Non-current assets	6,562,096,027	(3,913,984,446)	2,648,111,581	6,071,590,930	(3,947,716,748)	2,123,874,182
Investments in associates and joint ventures	5,008,234,005	(4,503,664,434)	504,569,571	4,944,356,290	(4,519,128,821)	425,227,469
Deferred tax assets	-	589,679,987	589,679,987	-	571,412,073	571,412,073
Total assets	<u>₩7,183,091,162</u>	<u>₩ (3,913,984,446)</u>	<u>₩3,269,106,716</u>	<u>₩7,533,016,106</u>	<u>₩ (3,947,716,748)</u>	<u>₩3,585,299,358</u>
Liabilities						
Current liabilities	₩2,287,442,106	₩ -	₩2,287,442,106	₩2,476,545,193	₩ -	₩2,476,545,193
Derivative liabilities	1,933,562,427	-	1,933,562,427	1,874,470,943	-	1,874,470,943
Non-current liabilities	919,198,164	(505,315,891)	413,882,273	974,091,453	(526,670,614)	447,420,839
Deferred tax liabilities	505,315,891	(505,315,891)	-	526,670,614	(526,670,614)	-
Total liabilities	<u>3,206,640,270</u>	<u>(505,315,891)</u>	<u>2,701,324,379</u>	<u>3,450,636,646</u>	<u>(526,670,614)</u>	<u>2,923,966,032</u>
Equity						
Share premium	2,487,313,082	9,313,769	2,496,626,851	2,487,313,082	9,313,769	2,496,626,851
Accumulated other comprehensive loss	(4,009,447)	(3,780,868)	(7,790,315)	4,947,005	(3,948,258)	998,747
Retained earnings	1,327,734,757	(3,414,201,456)	(2,086,466,699)	1,424,706,873	(3,426,411,645)	(2,001,704,772)
Total equity	<u>3,976,450,892</u>	<u>(3,408,668,555)</u>	<u>567,782,337</u>	<u>4,082,379,460</u>	<u>(3,421,046,134)</u>	<u>661,333,326</u>
Total liabilities and equity	<u>₩7,183,091,162</u>	<u>₩ (3,913,984,446)</u>	<u>₩3,269,106,716</u>	<u>₩7,533,016,106</u>	<u>₩ (3,947,716,748)</u>	<u>₩3,585,299,358</u>

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(iii) As at December 31, 2015 and 2014

(in thousands of Korean won)

	2015			2014		
	Before restatement	Adjustment	After restatement	Before restatement	Adjustment	After restatement
Assets						
Current assets	₩ 192,853,895	₩ -	₩ 192,853,895	₩ 216,838,130	₩ (106,686,799)	₩ 110,151,331
Non-current assets	5,767,639,821	(3,982,608,742)	1,785,031,079	1,138,859,825	(57,023,021)	1,081,836,804
Investments in associates and joint ventures	4,837,539,390	(4,533,780,087)	303,759,303	53,921,974	311,196,041	365,118,015
Deferred tax assets	-	551,171,345	551,171,345	-	-	-
Total assets	<u>₩5,960,493,716</u>	<u>₩ (3,982,608,742)</u>	<u>₩1,977,884,974</u>	<u>₩1,355,697,955</u>	<u>₩ (163,709,820)</u>	<u>₩1,191,988,135</u>
Liabilities						
Current liabilities	₩1,911,669,204	₩ -	₩1,911,669,204	₩ 236,793,886	₩ (86,282,991)	₩ 150,510,895
Derivative liabilities	1,820,439,050	-	1,820,439,050	-	-	-
Non-current liabilities	1,273,990,355	(581,574,424)	692,415,931	488,171,822	(52,359,864)	435,811,958
Deferred tax liabilities	581,574,424	(581,574,424)	-	-	-	-
Total liabilities	<u>3,185,659,559</u>	<u>(581,574,424)</u>	<u>2,604,085,135</u>	<u>724,965,708</u>	<u>(138,642,855)</u>	<u>586,322,853</u>
Equity						
Share premium	1,030,620,605	9,313,769	1,039,934,374	801,800,313	7,117,379	808,917,692
Accumulated other comprehensive loss	4,839,666	(3,778,548)	1,061,118	9,170,941	(203,819)	8,967,122
Retained earnings	1,601,530,283	(3,406,569,538)	(1,805,039,255)	(314,664,171)	611,394	(314,052,777)
Non-controlling interest	-	-	-	32,591,919	(32,591,919)	-
Total equity	<u>2,774,834,157</u>	<u>(3,401,034,318)</u>	<u>(626,200,161)</u>	<u>630,732,247</u>	<u>(25,066,965)</u>	<u>605,665,282</u>
Total liabilities and equity	<u>₩5,960,493,716</u>	<u>₩ (3,982,608,742)</u>	<u>₩1,977,884,974</u>	<u>₩1,355,697,955</u>	<u>₩ (163,709,820)</u>	<u>₩1,191,988,135</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
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(iv) As at December 31, 2013 and 2012

(in thousands of Korean won)

	2013			2012		
	Before restatement	Adjustment	After restatement	Before restatement	Adjustment	After restatement
Assets						
Current assets	₩ 156,395,179	₩ (112,364,693)	₩ 44,030,486	₩ 217,858,628	₩ (141,513,019)	₩ 76,345,609
Non-current assets	503,166,162	16,598,535	519,764,697	344,486,626	77,518,098	422,004,724
Investments in associates and joint ventures	-	166,631,898	166,631,898	-	163,615,991	163,615,991
Total assets	<u>₩ 659,561,341</u>	<u>₩ (95,766,158)</u>	<u>₩ 563,795,183</u>	<u>₩ 562,345,254</u>	<u>₩ (63,994,921)</u>	<u>₩ 498,350,333</u>
Liabilities						
Current liabilities	₩ 169,229,512	₩ (40,234,504)	₩ 128,995,008	₩ 107,516,998	₩ (34,243,680)	₩ 73,273,318
Non-current liabilities	105,868,478	(26,406,024)	79,462,454	6,366,172	(1,059,834)	5,306,338
Total liabilities	<u>275,097,990</u>	<u>(66,640,528)</u>	<u>208,457,462</u>	<u>113,883,170</u>	<u>(35,303,514)</u>	<u>78,579,656</u>
Equity						
Share premium	516,719,342	-	516,719,342	452,386,129	-	452,386,129
Accumulated other comprehensive loss	(2,341,862)	(136,268)	(2,478,130)	(990,977)	(97,997)	(1,088,974)
Retained earnings	(218,754,883)	416,268	(218,338,615)	(83,168,967)	280,000	(82,888,967)
Non-controlling interest	29,405,629	(29,405,629)	-	28,873,410	(28,873,410)	-
Total equity	<u>384,463,351</u>	<u>(29,125,630)</u>	<u>355,337,721</u>	<u>448,462,084</u>	<u>(28,691,407)</u>	<u>419,770,677</u>
Total liabilities and equity	<u>₩ 659,561,341</u>	<u>₩ (95,766,158)</u>	<u>₩ 563,795,183</u>	<u>₩ 562,345,254</u>	<u>₩ (63,994,921)</u>	<u>₩ 498,350,333</u>

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(b) Impact on Financial Performance

If the Company loses the administrative litigation, it shall restate the statements of comprehensive income (consolidated statements of comprehensive income from 2012 to 2014) as indicated by the Securities and Futures Commission and its impact shall be as follows.

(i) For the years ended December 31, 2019 and 2018

(in thousands of Korean won)

	2019			2018		
	Before restatement	Adjustment	After restatement	Before restatement	Adjustment	After restatement
Revenue	₩ 701,591,860	₩ -	₩ 701,591,860	₩ 535,805,801	₩ -	₩ 535,805,801
Cost of revenue	(509,584,425)	-	(509,584,425)	(390,645,859)	-	(390,645,859)
Gross profit	192,007,435	-	192,007,435	145,159,942	-	145,159,942
Operating profit	91,742,387	-	91,742,387	55,678,745	-	55,678,745
Equity gain (loss) on investments in associates and joint ventures	72,940,116	17,981,780	90,921,896	(129,337,333)	(3,364,228)	(132,701,561)
Earnings from disposal of investment equity in associates	-	-	-	389,213,672	2,119,447,933	2,508,661,605
Net profit before Income tax	155,435,309	17,981,780	173,417,089	302,975,837	2,116,083,704	2,419,059,541
Income tax expenses (benefit)	(47,468,790)	7,691,429	(39,777,361)	78,866,642	513,831,346	592,697,988
Net profit for the year	202,904,099	10,290,351	213,194,450	224,109,195	1,602,252,358	1,826,361,553
Other comprehensive loss	(3,918,805)	-	(3,918,805)	(2,966,514)	(274,831)	(3,241,345)
Total comprehensive income for the year	₩ 198,985,294	₩ 10,290,351	₩ 209,275,645	₩ 221,142,681	₩1,601,977,527	₩1,823,120,208

Samsung Biologics Co., Ltd.
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(ii) For the years ended December 31, 2017 and 2016

(in thousands of Korean won)

	2017			2016		
	Before restatement	Adjustment	After restatement	Before restatement	Adjustment	After restatement
Revenue	₩ 464,629,334	₩ -	₩ 464,629,334	₩ 294,622,021	₩ -	₩ 294,622,021
Cost of revenue	(332,829,273)	-	(332,829,273)	(268,088,358)	-	(268,088,358)
Gross profit	131,800,061	-	131,800,061	26,533,663	-	26,533,663
Operating profit (loss)	65,979,402	-	65,979,402	(30,422,639)	-	(30,422,639)
Equity gain (loss) on investments in associates and joint ventures	(129,602,338)	15,464,387	(114,137,951)	(129,222,650)	14,651,266	(114,571,384)
Losses on valuation of derivatives	(59,091,484)	-	(59,091,484)	(54,031,894)	-	(54,031,894)
Net profit (loss) before Income tax	(115,994,753)	15,464,387	(100,530,366)	(230,900,983)	14,651,266	(216,249,717)
Income tax expenses (benefit)	(19,022,637)	3,254,198	(15,768,439)	(54,077,574)	34,493,373	(19,584,201)
Net profit (loss) for the year	(96,972,116)	12,210,189	(84,761,927)	(176,823,409)	(19,842,107)	(196,665,516)
Other comprehensive income (loss)	(8,956,451)	167,389	(8,789,062)	107,338	(169,709)	(62,371)
Total comprehensive income (loss) for the year	₩ (105,928,567)	₩ 12,377,578	₩ (93,550,989)	₩ (176,716,071)	₩ (20,011,816)	₩ (196,727,887)

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(iii) For the years ended December 31, 2015 and 2014

(in thousands of Korean won)

	2015			2014		
	Before restatement	Adjustment	After restatement	Before restatement	Adjustment	After restatement
Revenue	₩ 91,278,029	₩ (23,901,796)	₩ 67,376,233	₩ 105,148,784	₩ (76,130,543)	₩ 29,018,241
Cost of revenue	(114,784,617)	48,453,429	(66,331,188)	(124,868,406)	41,530,645	(83,337,761)
Gross profit	(23,506,588)	24,551,633	1,045,045	(19,719,622)	(34,599,898)	(54,319,520)
Operating profit (loss)	(203,642,444)	160,536,379	(43,106,065)	(119,541,186)	38,655,238	(80,885,948)
Equity loss on investments in associates and joint ventures	(19,395,735)	(151,241,000)	(170,636,735)	(32,679,484)	(35,067,057)	(67,746,541)
Earnings (losses) from disposal of investment equity in subsidiary	4,543,610,535	(4,543,610,535)	-	-	-	-
Losses on valuation of derivatives	(1,820,439,050)	-	(1,820,439,050)	-	-	-
Net profit (loss) before Income tax	2,488,062,610	(4,528,216,321)	(2,040,153,711)	(99,688,402)	3,974,241	(95,714,161)
Income tax expenses (benefit)	583,116,693	(1,132,283,925)	(549,167,232)	-	-	-
Net profit (loss) for the year	1,904,945,917	(3,395,932,396)	(1,490,986,479)	(99,688,402)	3,974,241	(95,714,161)
Other comprehensive income (loss)	(8,535,494)	629,489	(7,906,005)	11,295,735	149,517	11,445,252
Total comprehensive income (loss) for the year	<u>₩1,896,410,423</u>	<u>₩ (3,395,302,907)</u>	<u>₩ (1,498,892,484)</u>	<u>₩ (88,392,667)</u>	<u>₩ 4,123,758</u>	<u>₩ (84,268,909)</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
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(iii) For the years ended December 31, 2013 and 2012

(in thousands of Korean won)

	2013			2012		
	Before restatement	Adjustment	After restatement	Before restatement	Adjustment	After restatement
Revenue	₩ 17,120,845	₩ (17,077,340)	₩ 43,505	₩ -	₩ -	₩ -
Cost of revenue	(54,438,932)	54,399,778	(39,154)	-	-	-
Gross profit	(37,318,087)	37,322,438	4,351	-	-	-
Operating profit (loss)	(154,092,746)	89,212,838	(64,879,908)	(91,357,686)	51,645,812	(39,711,874)
Equity loss on investments in associates and joint ventures	-	(73,031,096)	(73,031,096)	-	(40,016,257)	(40,016,257)
Net profit (loss) before Income tax	(148,473,756)	13,024,108	(135,449,648)	(80,241,939)	7,341,692	(72,900,247)
Income tax expenses	-	-	-	2,066,460	-	2,066,460
Net profit (loss) for the year	(148,473,756)	13,024,108	(135,449,648)	(82,308,399)	7,341,692	(74,966,707)
Other comprehensive income (loss)	(1,424,072)	34,915	(1,389,157)	(1,038,989)	(50,393)	(1,089,382)
Total comprehensive income (loss) for the year	₩ (149,897,828)	₩ 13,059,023	₩ (136,838,805)	₩ (83,347,388)	₩ 7,291,299	₩ (76,056,089)

36. Events After the Reporting Period

On January 29, 2020, in order to expand stable orders and overseas operations for CDO business, the Company established Samsung Biologics America, Inc., a subsidiary, in the United States(Percentage of ownership: 100%).

**Report on Independent Auditor's
Audit of Internal Control over Financial Reporting**

(English Translation of a Report Originally Issued in Korean)

To Shareholders and the Board of Directors of
Samsung Biologics Co., Ltd.

Opinion on Internal Control over Financial Reporting

We have audited Samsung Biologics Co., Ltd.'s (the "Company") Internal Control over Financial Reporting as of December 31, 2019, based on *Conceptual Framework for Designing and Operating Internal Control over Financial Reporting*.

In our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2019, based on *Conceptual Framework for Designing and Operating Internal Control over Financial Reporting*.

We also have audited, in accordance with Korean Standards on Auditing, the financial statements of the Company, which comprise the statement of financial position as at December 31, 2019, and the statement of comprehensive income, statement of changes in equity and statement of cash flow for the year then ended, and notes to the financial statements including a summary of significant accounting policies, and our report dated March 5, 2020 expressed an unqualified opinion.

Basis for Opinion on Internal Control over Financial Reporting

We conducted our audit in accordance with Korean Standards on Auditing. Our responsibility under these standards are further described in the *Auditor's Responsibilities for the Audit of the Internal Control over Financial Reporting* section of our report. We are independent of the Company in accordance with the ethical requirements of the Republic of Korea that are relevant to our audit of internal control over financial reporting and we have fulfilled our other ethical responsibilities in accordance with these requirements. We believe that the audit evidence we have obtained is sufficient and appropriate to provide a basis for our audit opinion.

Responsibilities of Management and Those Charged with Governance for Internal Control over Financial Reporting

Management is responsible for designing, implementing and maintaining effective internal control over financial reporting, and for its assessment about the effectiveness of internal control over financial reporting, included in the accompanying Report on the Effectiveness of Internal Control over Financial Reporting.

Those charged with governance have the responsibilities for overseeing internal control over financial reporting.

Auditor's Responsibilities for the Audit of Internal Control over Financial Reporting

Our responsibility is to express opinion on the Company's internal control over financial reporting based on our audit. We conducted the audit in accordance with Standards on Auditing. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects.

An audit of internal control over financial reporting involves performing procedures to obtain audit evidence about whether a material weakness exists. The procedures selected depend on the auditor's judgment, including the assessment of the risks that a material weakness exists. An audit includes obtaining an understanding of internal control over financial reporting and testing and evaluating the design and operating effectiveness of internal control over financial reporting based on the assessed risk.

Definition and Inherent Limitations of Internal Control over Financial Reporting

An entity's internal control over financial reporting is a process effected by those charged with governance, management, and other personnel, designed to provide reasonable assurance regarding the preparation of reliable financial statements in accordance with International Financial Reporting Standards as adopted by the Republic of Korea. An entity's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the entity; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with International Financial Reporting Standards as adopted by the Republic of Korea, and that receipts and expenditures of the entity are being made only in accordance with authorizations of management and those charged with governance; and (3) provide reasonable assurance regarding prevention, or timely detection and correction of unauthorized acquisition, use, or disposition of the entity's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent, or detect and correct, misstatements. Also, projections of any assessment of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Other matter

As described in the attached Report on the Effectiveness of Internal Control over Financial Reporting, the Company's management identified material weaknesses during 2019 which occurred during 2018. The material weaknesses were the deficiency in reviewing certain underlying data used in preparation of the statements of cash flows and the deficiency in entity-level controls regarding the protection of information and data from document destruction and concealment. Management concluded that the material weaknesses were remediated as of December 31, 2019.

The engagement partner on the audit resulting in this independent auditor's report is Joonki Hong, Certified Public Accountant.

Samil PricewaterhouseCoopers

Seoul, Korea

March 5, 2020

This report is effective as of March 5, 2020, the audit report date. Certain subsequent events or circumstances, which may occur between the audit report date and the time of reading this report, could have a material impact on the Company's internal control over financial reporting thereto. Accordingly, the readers of the audit report should understand that there is a possibility that the above audit report may have to be revised to reflect the impact of such subsequent events or circumstances, if any.

Report on the Effectiveness of Internal Control over Financial Reporting

(English Translation of a Report Originally Issued in Korean)

To the Shareholders, Board of Directors and Audit Committee of
Samsung Biologics Co., Ltd.

We, as the Chief Executive Officer (CEO) and the Internal Control over Financial Reporting Officer of Samsung Biologics Co., Ltd. (the "Company"), assessed the effectiveness of the design and operation of the Company's Internal Control over Financial Reporting (ICFR) for the year ended December 31, 2019.

The Company's management, including ourselves, is responsible for designing and operating ICFR. We assessed the design and operating effectiveness of ICFR in the prevention and detection of an error or fraud which may cause material misstatements in the preparation and disclosure of reliable financial statements. We designed and operated ICFR in accordance with *Conceptual Framework for Designing and Operating Internal Control over Financial Reporting* established by the Operating Committee of Internal Control over Financial Reporting in Korea (the ICFR Committee). And, we conducted an evaluation of ICFR based on *Best Practice Guideline for Evaluating and Reporting Effectiveness of Internal Control over Financial Reporting* established by the ICFR Committee.

Based on the assessment results, we believe that the Company's ICFR, as of December 31, 2019, is designed and operating effectively, in all material respects, in accordance with *Conceptual Framework for Designing and Operating Internal Control over Financial Reporting*.

The following material weaknesses were identified during the current year. However, the material weaknesses have been remediated by establishing and implementing appropriate remediation plans within the current year.

Description of material weaknesses

- a. Deficiencies in entity-level controls, information and data protection policies, and related control activities regarding document destruction and concealment case occurred during 2018.
- b. Misstatements in preparing the statements of cash flows for the year ended December 31, 2018.

Remediation plans for material weaknesses

- a. Establishing an independent investigation committee and performing an investigation of these matters, enhancement and training of the code of ethics, strengthening of the oversight authority of the Audit Committee, and enhancement and inspection of information protection policies.
- b. Adding controls and strengthening control activities to assess the completeness and accuracy of the underlying data for preparing the statements of cash flows.

We certify that this report does not contain any untrue statement of a fact, or omit to state a fact necessary to be presented herein. We also certify that this report does not contain or present any statement which cause material misunderstandings, and we have reviewed and verified this report with sufficient due care.

Attachment

- Description of material weaknesses discovered during the current year and remediation plans

February 28, 2020

Tea Han Kim,
Chief Executive Officer

Dong Joong Kim,
Internal Control over Financial Reporting Officer

Attachment. Description of material weaknesses discovered during the current year and remediation

(a) Deficiencies in entity-level controls, information and data protection policies, and related control activities regarding document destruction and concealment case occurred during 2018.

The Company did not design and operate sufficient and appropriate control procedures in relation to 1) establishing an organizational structure and reporting system to protect information and data and prevent forgery, tampering, corruption and destruction, 2) establishing and operating related policies and procedures, and 3) establishing a monitoring system to identify and take actions against violations of the internal control system. This represents a material weakness in control environment, information and communication, and monitoring activities in relation to recording and retaining accounting records for the Company's internal and external reporting.

The Company took the following measures in 2019 to remediate this material weaknesses related to the control environment, information and communication, and monitoring activities.

(i) The Audit Committee formed an independent committee (that engaged an external expert) and independently conducted an investigation of the administrative lawsuit and prosecution investigation pursuant to the SFC's measures described in Note 35 to the financial statements. The following matters were investigated:

- Whether there was any potential illegal acts or inappropriate accounting other than what was identified in the court ruling.
- Whether there was inappropriate modification or deletion of journal entries and supporting documents in the Company's financial and other related systems and whether continuity in accounting information was retained.
- Evaluated the effect of the evidence presented in the trial on the existing accounting determinations, judgements of management, and judgements of those charged with governance.
- Determined if there were violations of internal policies, and assessment of the internal control deficiencies identified and the related remediation plans.

(ii) The code of ethics was enhanced to respond to inappropriate instructions from outside, and the CEO emphasized the need for ethical management to employees through internal communication.

(iii) All employees mandatorily received trainings on internal control over financial reporting and fraud prevention and pledged to comply with the Company's code of ethics and the law. Through these trainings, the Company specifically emphasized the importance of the whistleblower hotline, through which employees at all levels can anonymously submit or express concerns regarding accounting, financial reporting, and violations of the Company's code of ethics and the law.

(iv) Internal audit department was reorganized to directly report to the Audit Committee for the purpose of reinforcing the independent oversight authority of the Audit Committee.

(v) A Chief Information Security Officer was appointed to protect all information and data of the Company, including accounting information, and the Audit Committee performed monitoring procedures after enhancing the information and data protection policies.

(b) Misstatements in preparing statements of cash flows

The Company did not design and operate sufficient and appropriate control procedures in reviewing the completeness and accuracy of the data regarding the acquisition of property, plant and equipment. This deficiency represents material weakness in which cash flows from investing activities and cash flows from operating activities related to the acquisition of property, plant and equipment may not be properly presented.

The Company redesigned and operated comprehensive review procedures to verify the completeness and accuracy of the data regarding investment activities and to include multiple reviewers in the workflow to reinforce the completeness and accuracy of the statements of cash flows amounts and presentation.